



(12) **United States Patent**
Brooks et al.

(10) **Patent No.:** **US 12,403,026 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **VALVE WITH DOCKING STATION FOR GASTROINTESTINAL BALLOON**

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(73) Assignee: **Spatz FGIA, Inc.**, Fort Lauderdale, FL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **16/723,446**

(22) Filed: **Dec. 20, 2019**

(65) **Prior Publication Data**

US 2020/0197206 A1 Jun. 25, 2020

Related U.S. Application Data

(60) Provisional application No. 62/784,106, filed on Dec. 21, 2018.

(51) **Int. Cl.**
A61F 5/00 (2006.01)

(52) **U.S. Cl.**
CPC **A61F 5/0043** (2013.01); **A61F 5/003** (2013.01)

(58) **Field of Classification Search**

CPC A61F 2/0027; A61F 5/003; A61F 5/0056; A61F 2/004; A61F 6/16; A61F 2002/501; A61F 5/005; A61F 5/0036-0043; A61F 5/0073; A61M 25/1025; A61M 2039/1063; A61M 2039/1072; A61M 2039/2426; A61M 2039/267; A61M 25/10185; A61M 25/1054; A61M 39/06; A61M 39/0606-0613; A61M 39/0693;
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Primary Examiner — Elizabeth Houston

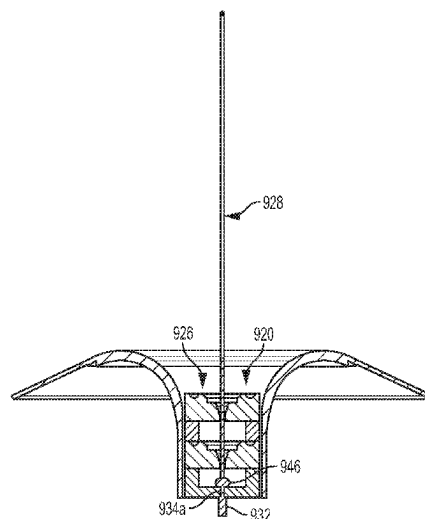
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(57) **ABSTRACT**

A gastrointestinal balloon system includes a gastrointestinal balloon and a valve located within the gastrointestinal balloon. The valve is configured to allow fluid flow into and out of the gastrointestinal balloon. The valve includes a string coupled to the valve, the string having a loop on a proximal end and a catheter having a lumen. The lumen is configured to receive a grasping tool. The grasping tool extends out a distal end of the catheter for grabbing the loop to align the catheter with the valve. The valve may be a series of self-sealing two-way valves, a check valve, or two-way valve. A method of using a valve in a gastrointestinal balloon is also described.

19 Claims, 30 Drawing Sheets



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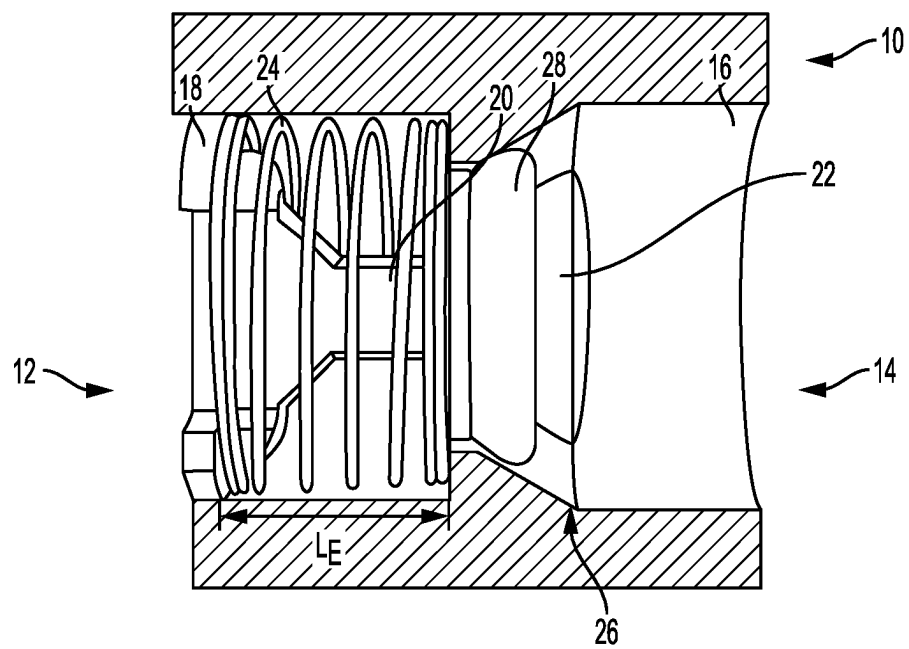


FIG. 1

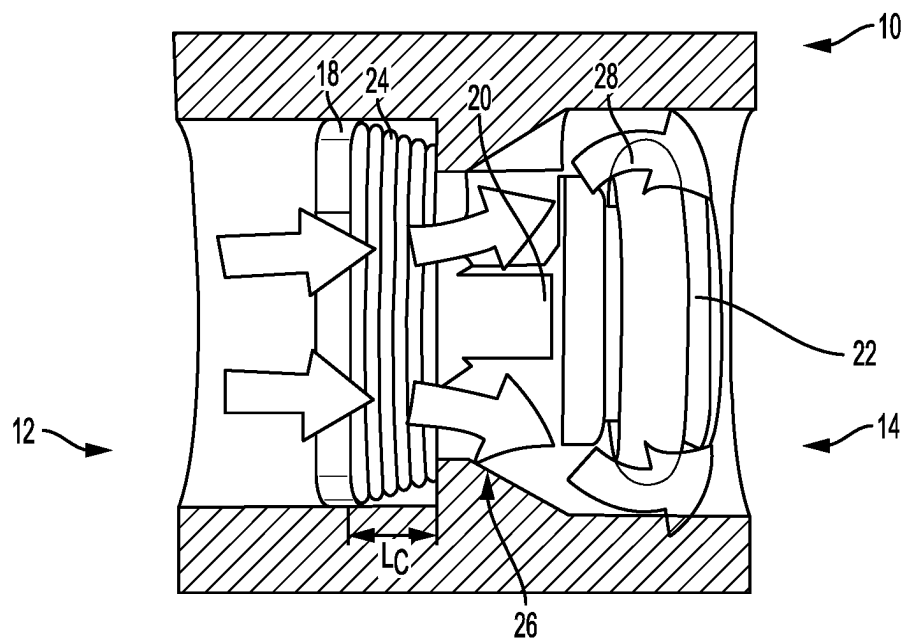


FIG. 2

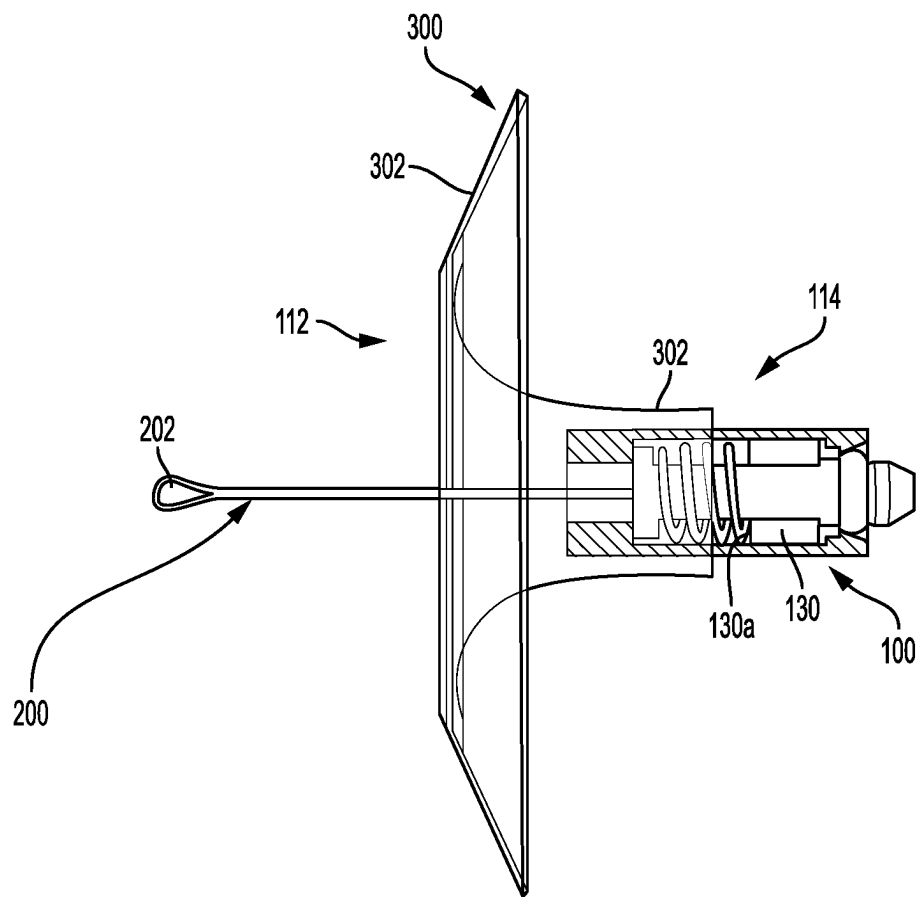


FIG. 3

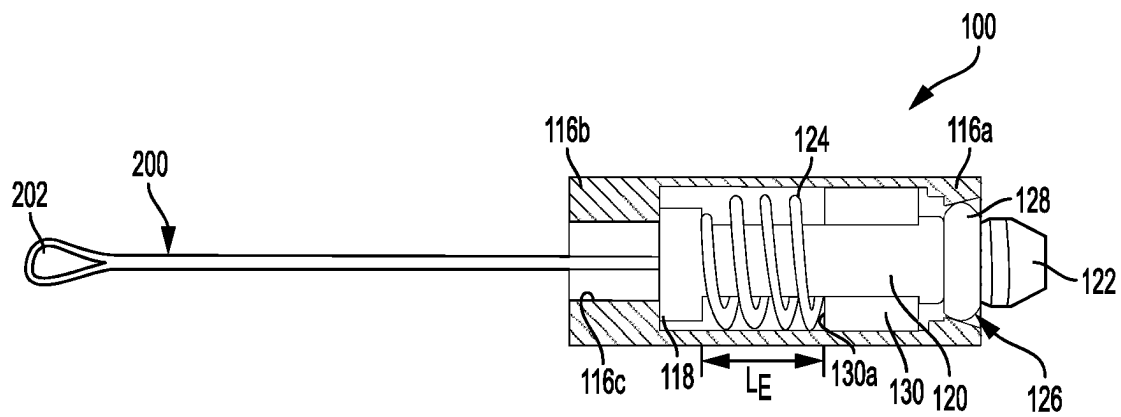


FIG. 4

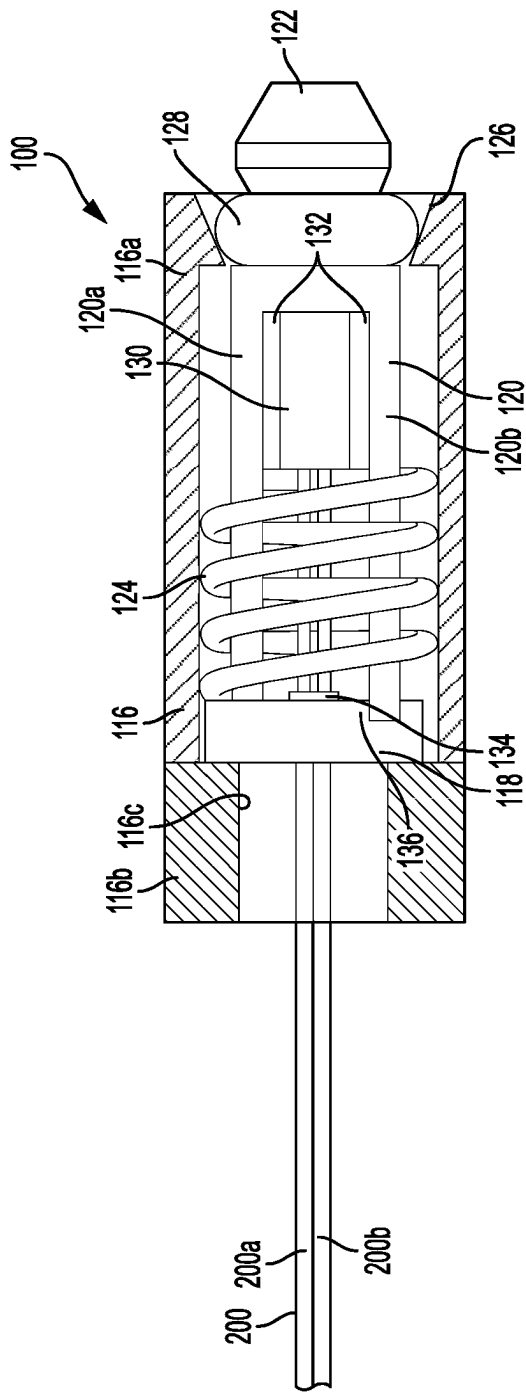


FIG. 5

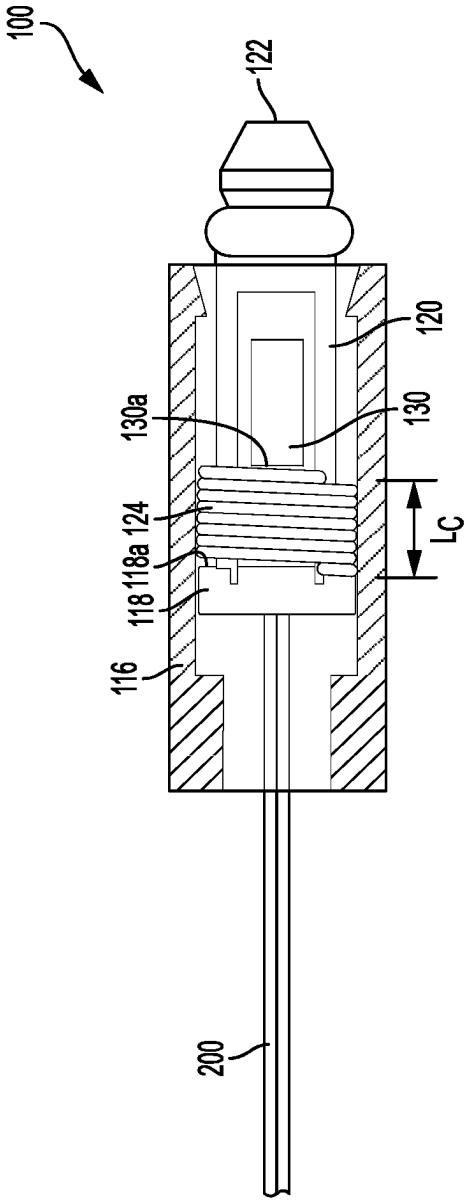


FIG. 6

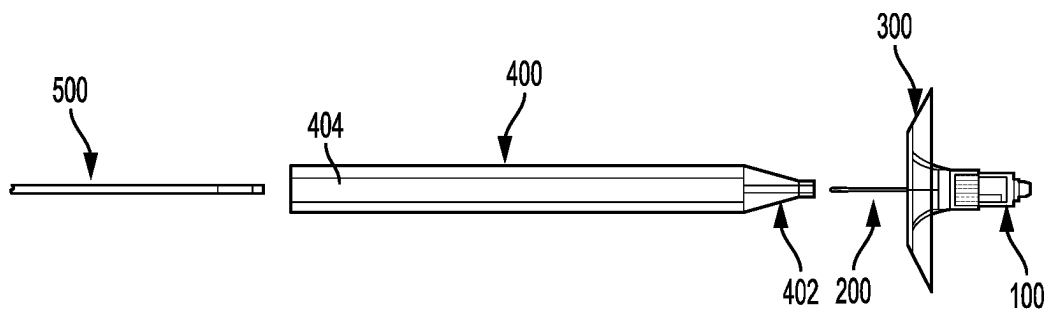


FIG. 7

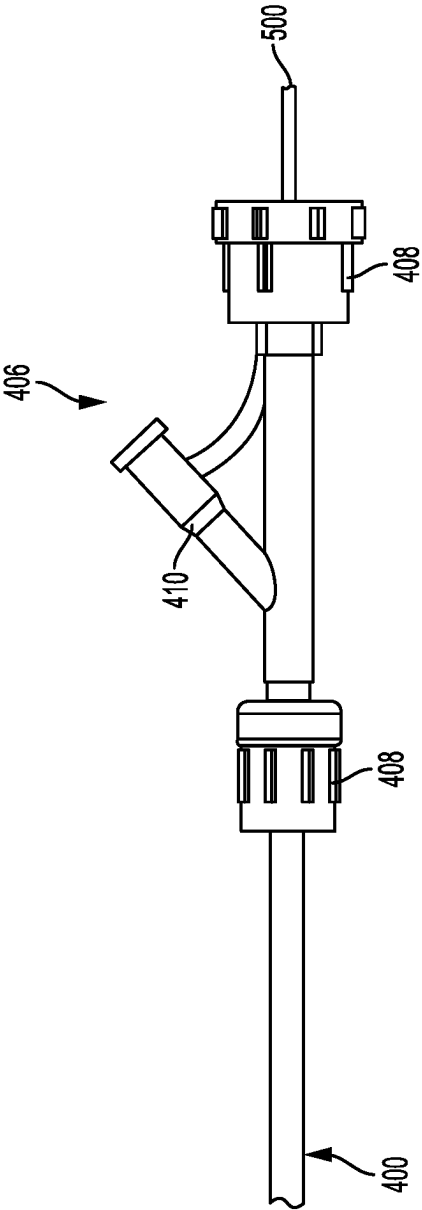


FIG. 8

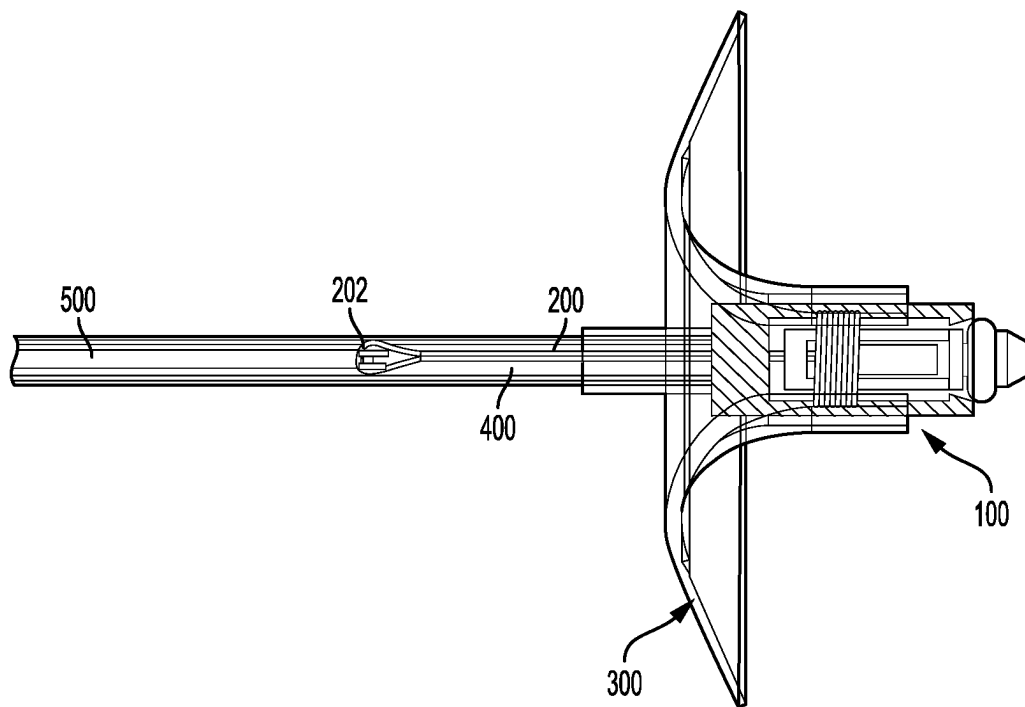


FIG. 9

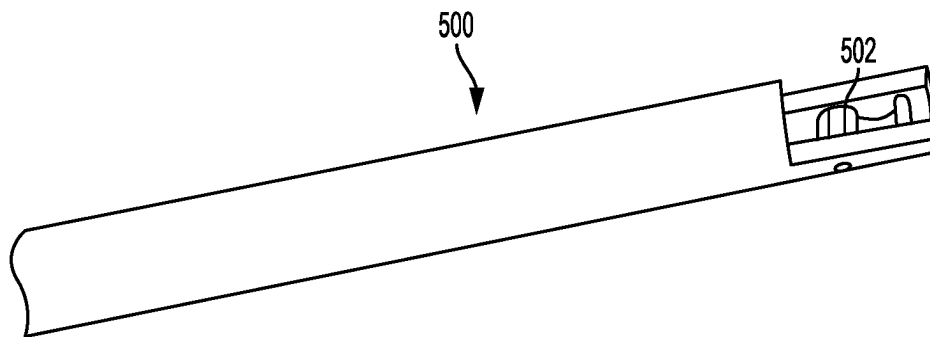


FIG. 10A

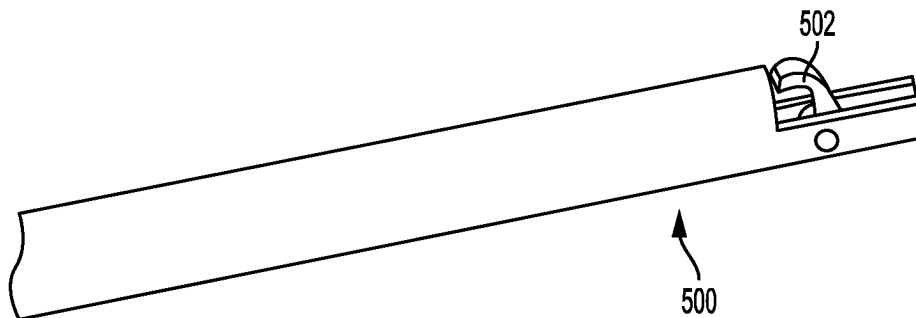


FIG. 10B

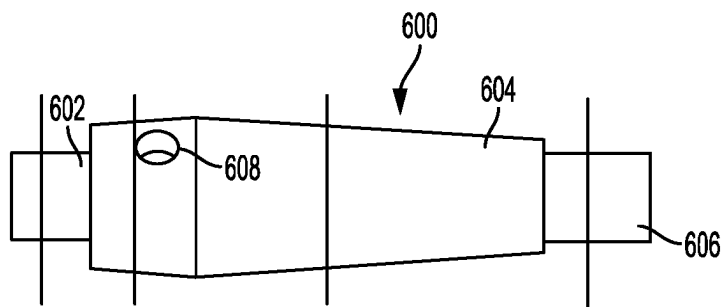


FIG. 11

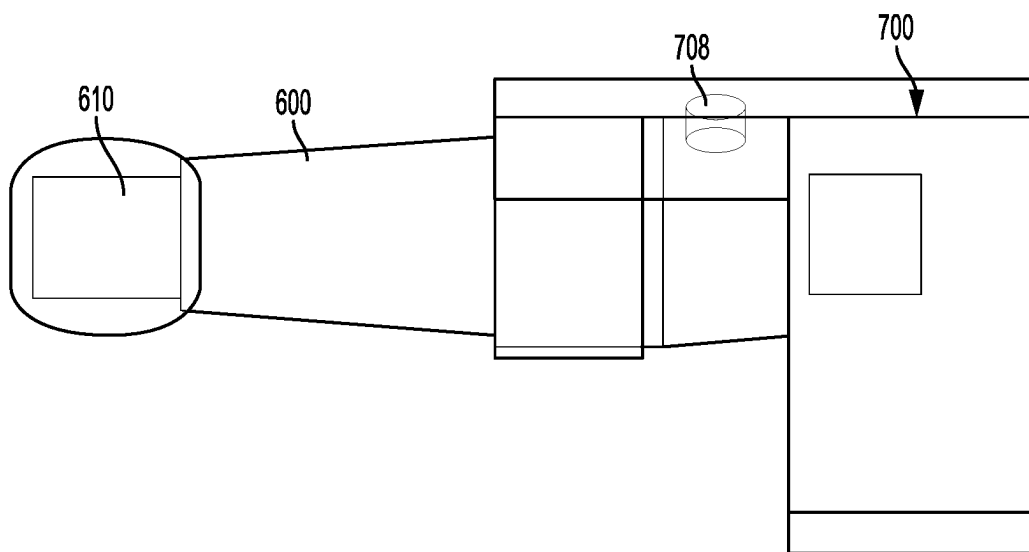


FIG. 12

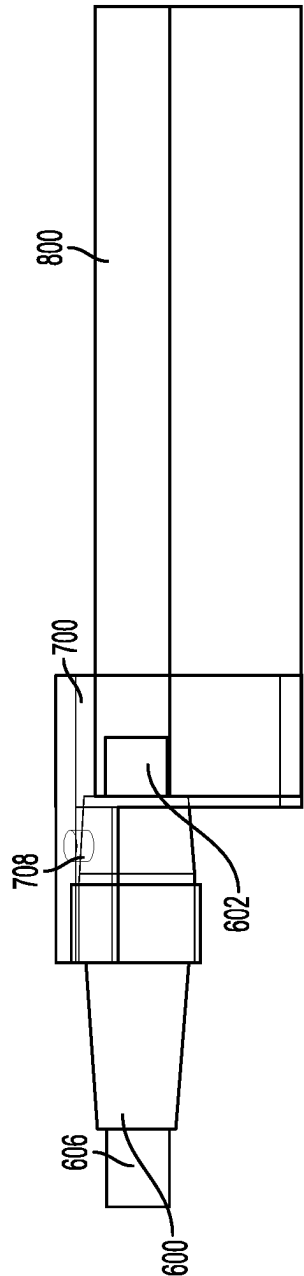


FIG. 13

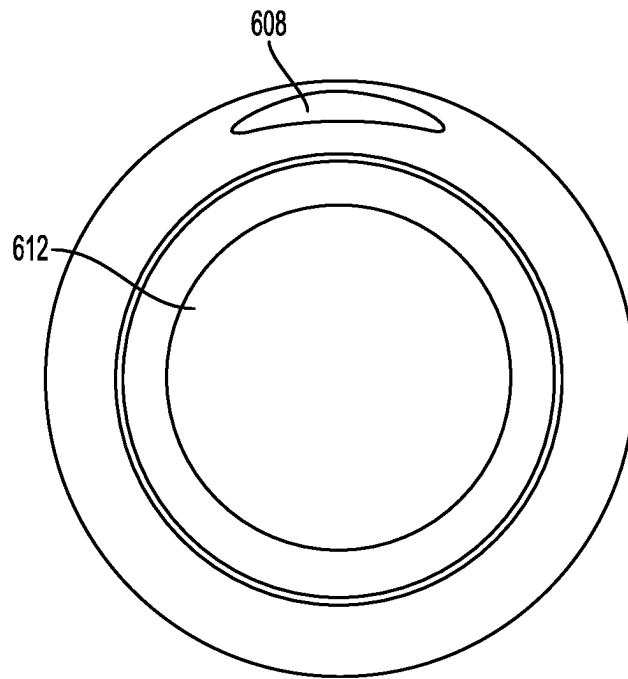


FIG. 14A

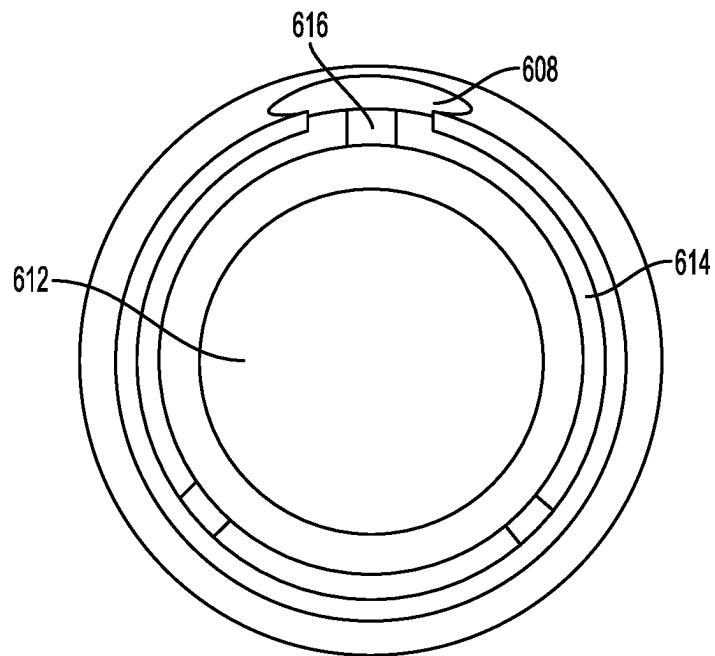


FIG. 14B

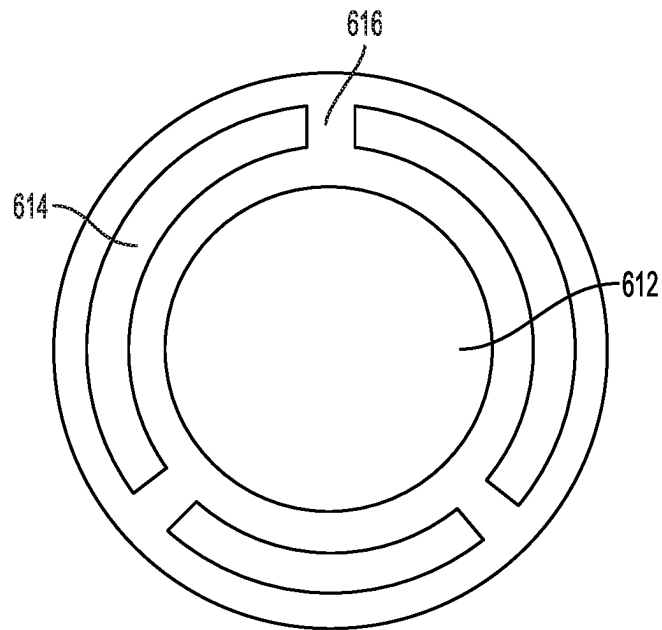


FIG. 14C

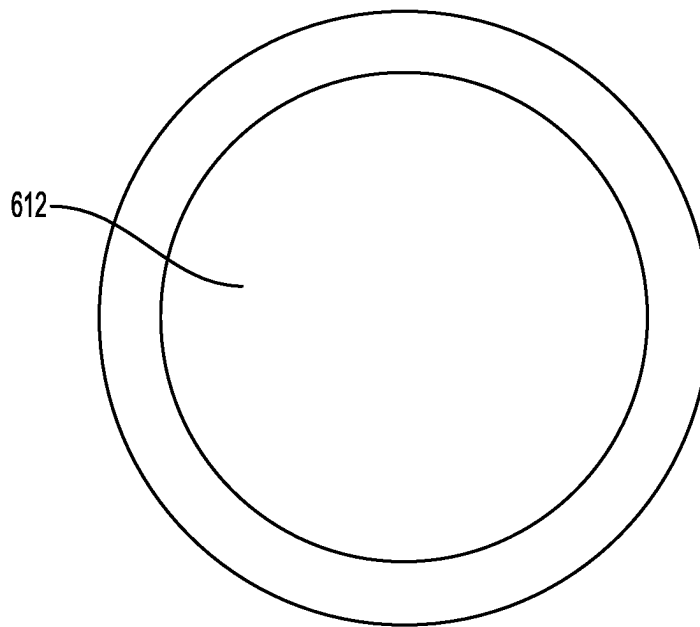


FIG. 14D

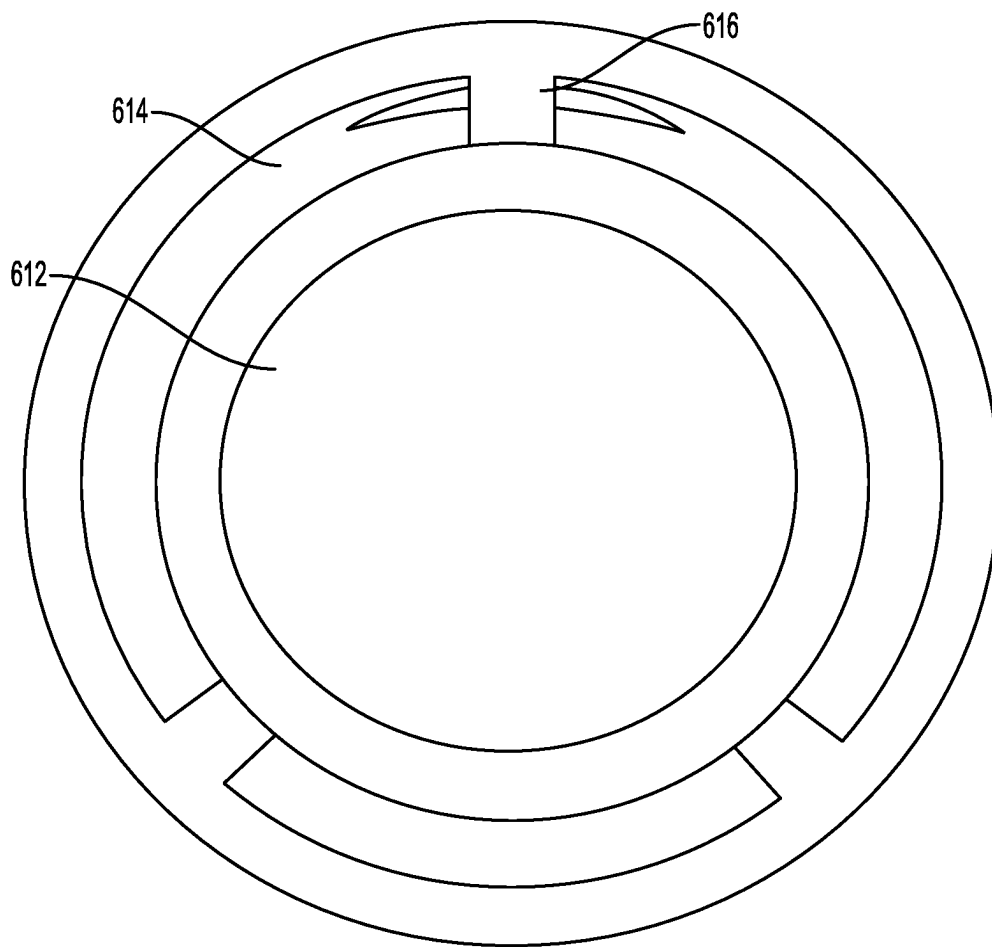


FIG. 14E

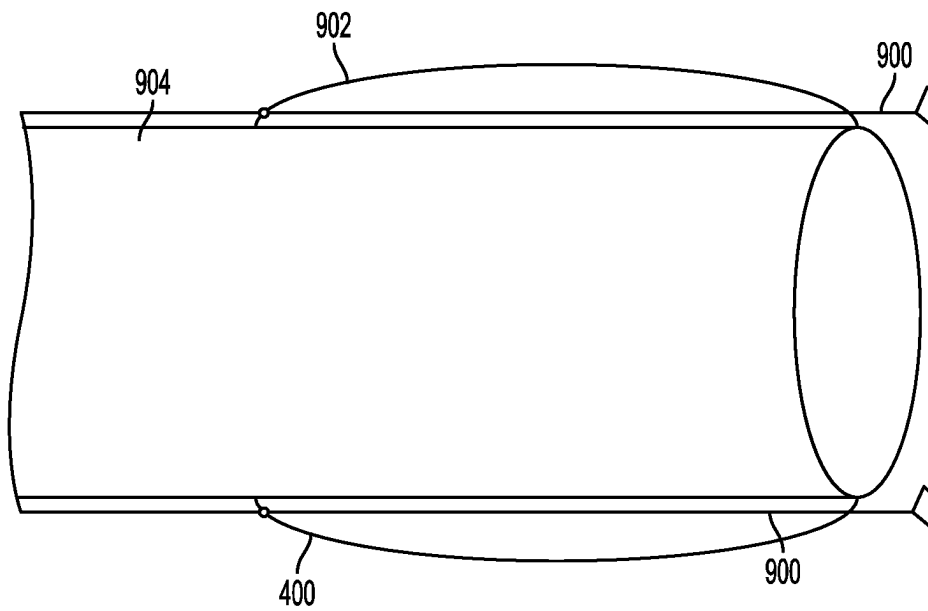


FIG. 15

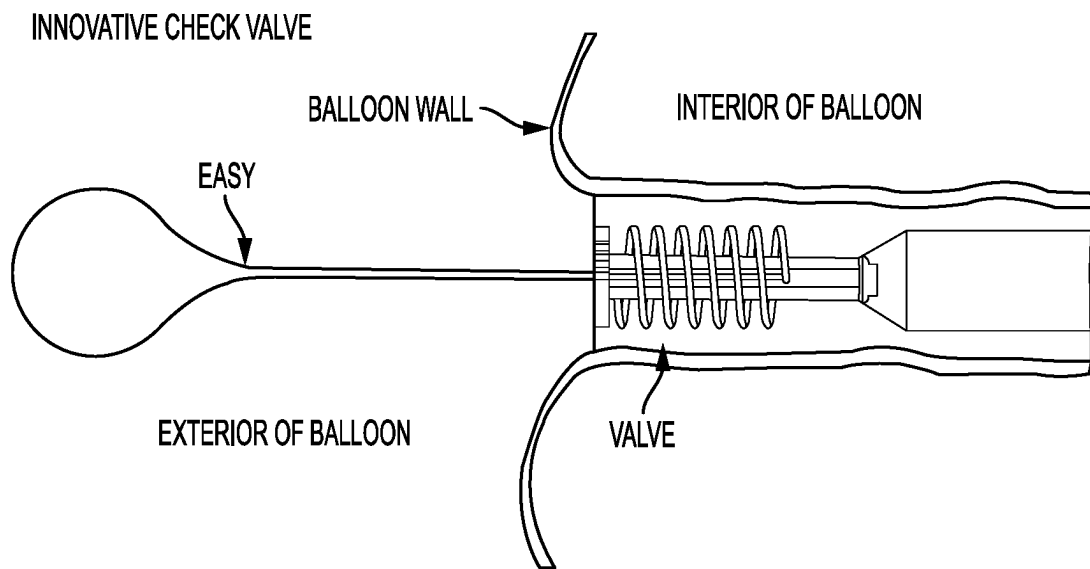


FIG. 16

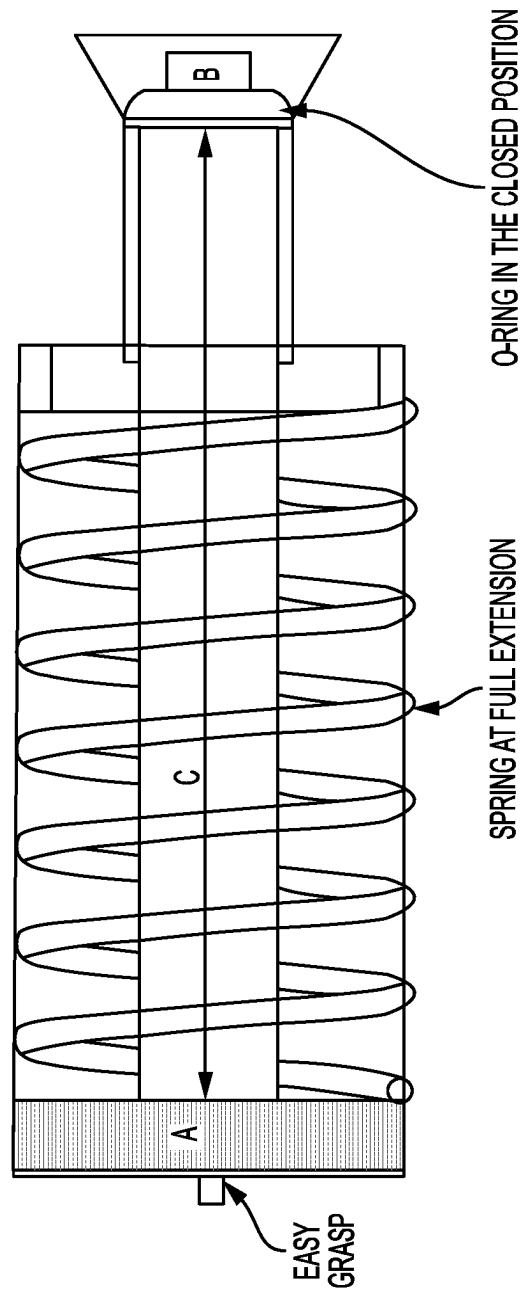


FIG. 17

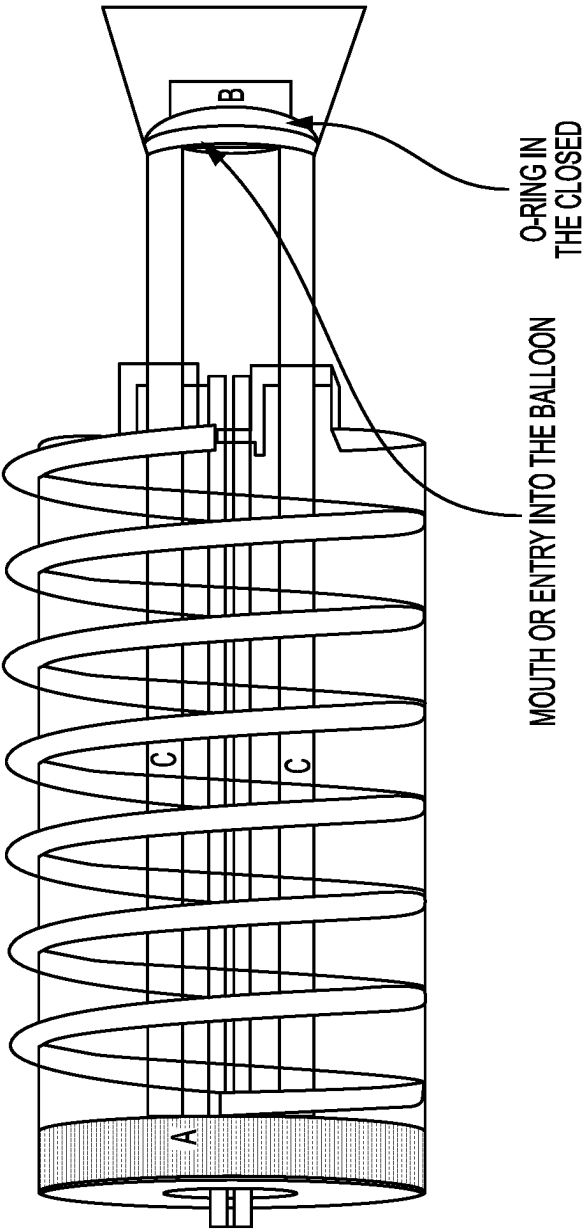


FIG. 18

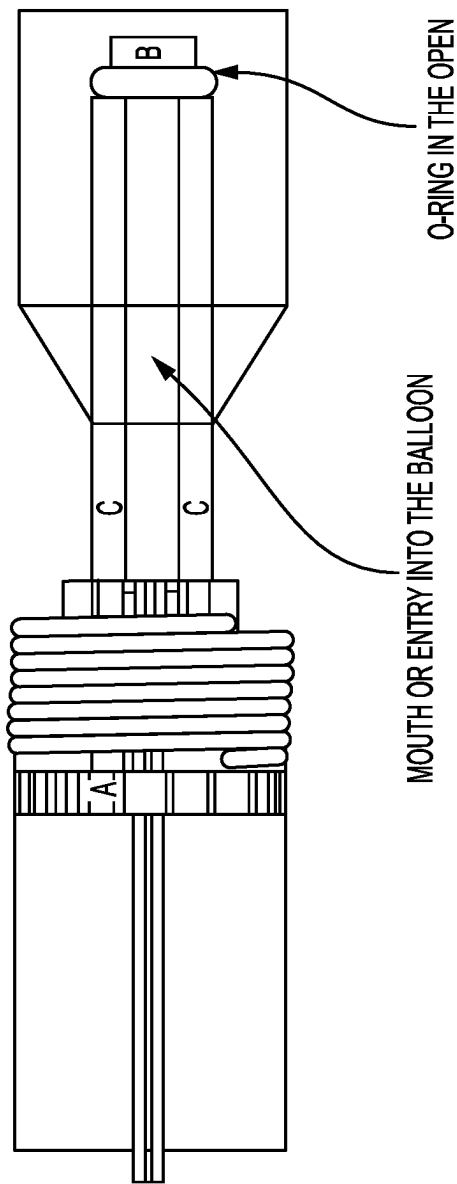


FIG. 19

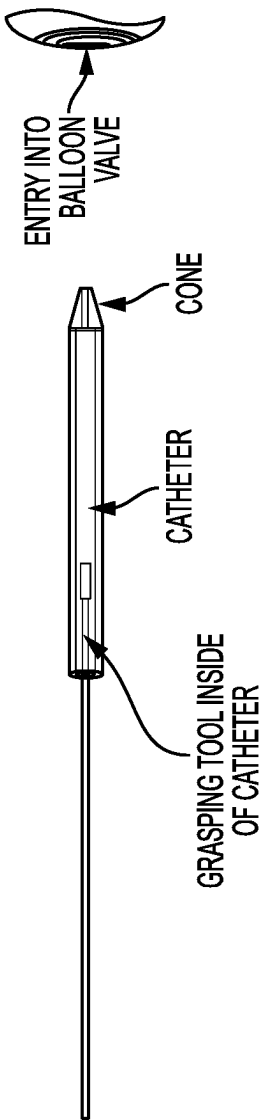


FIG. 20

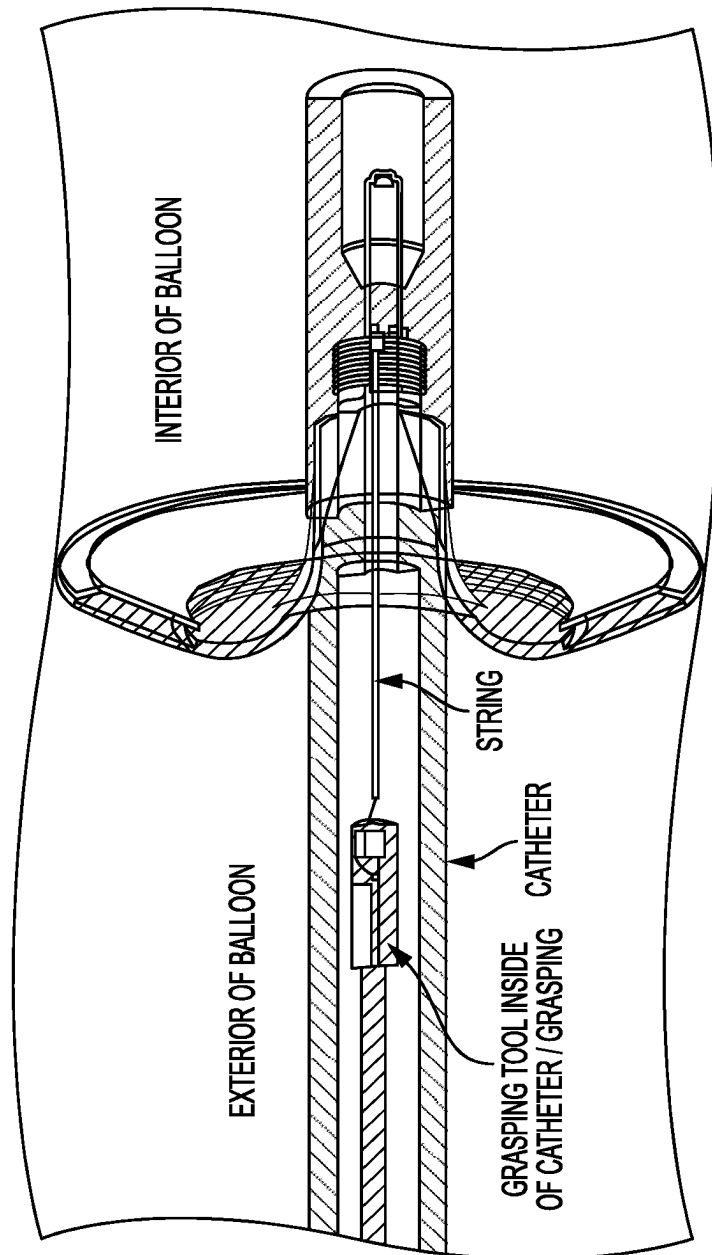


FIG. 21

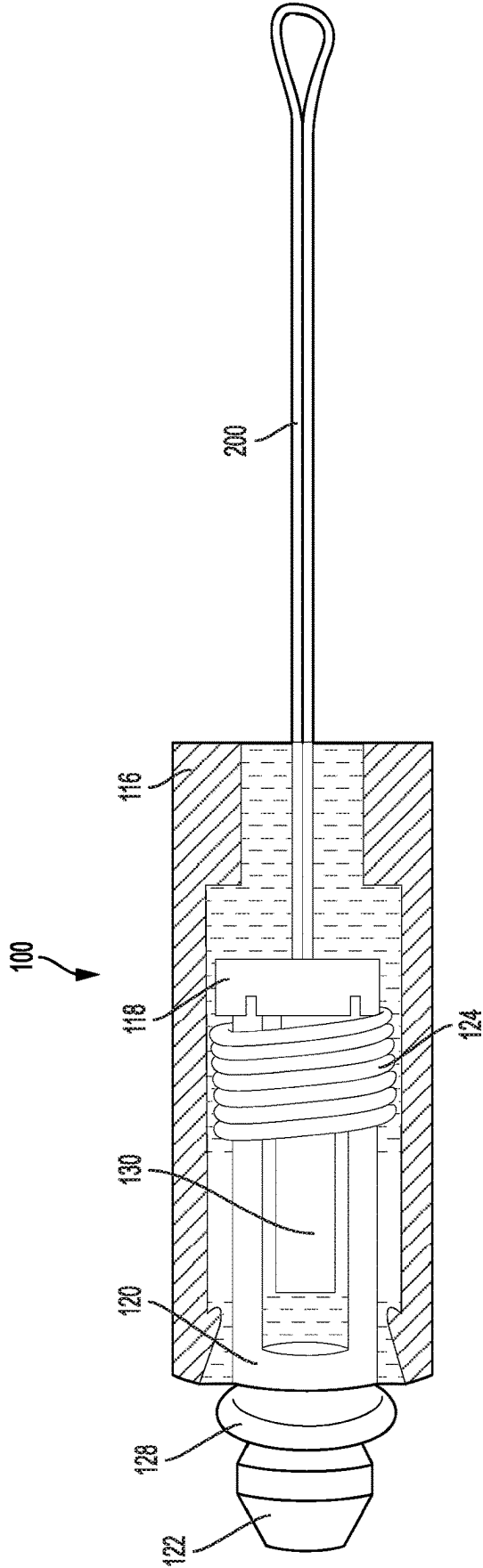


FIG. 22

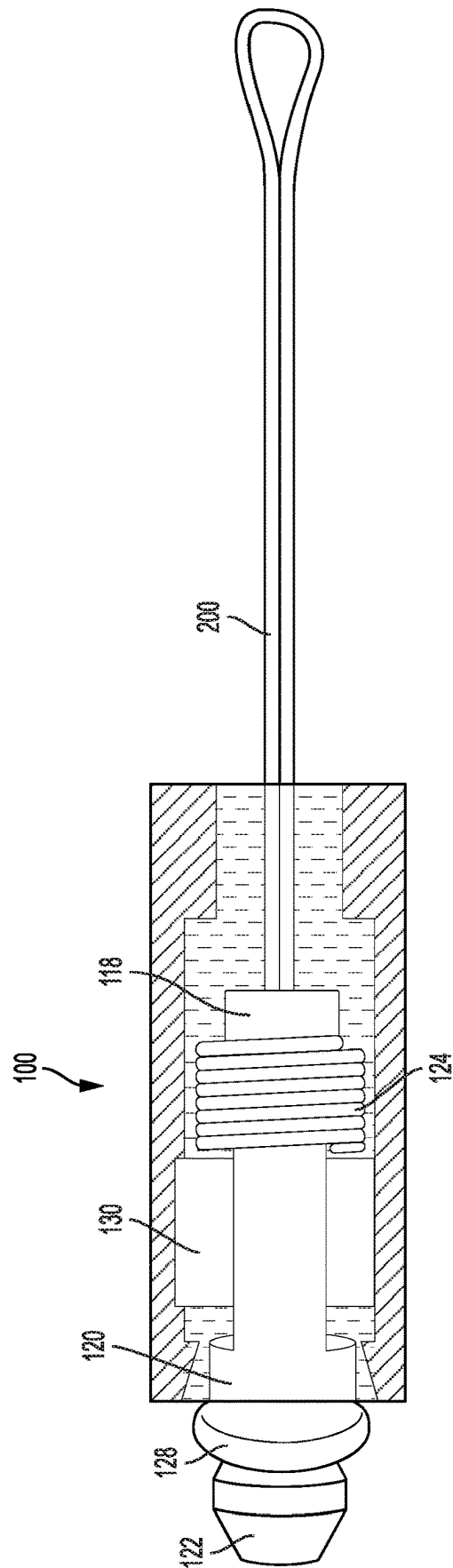


FIG. 23

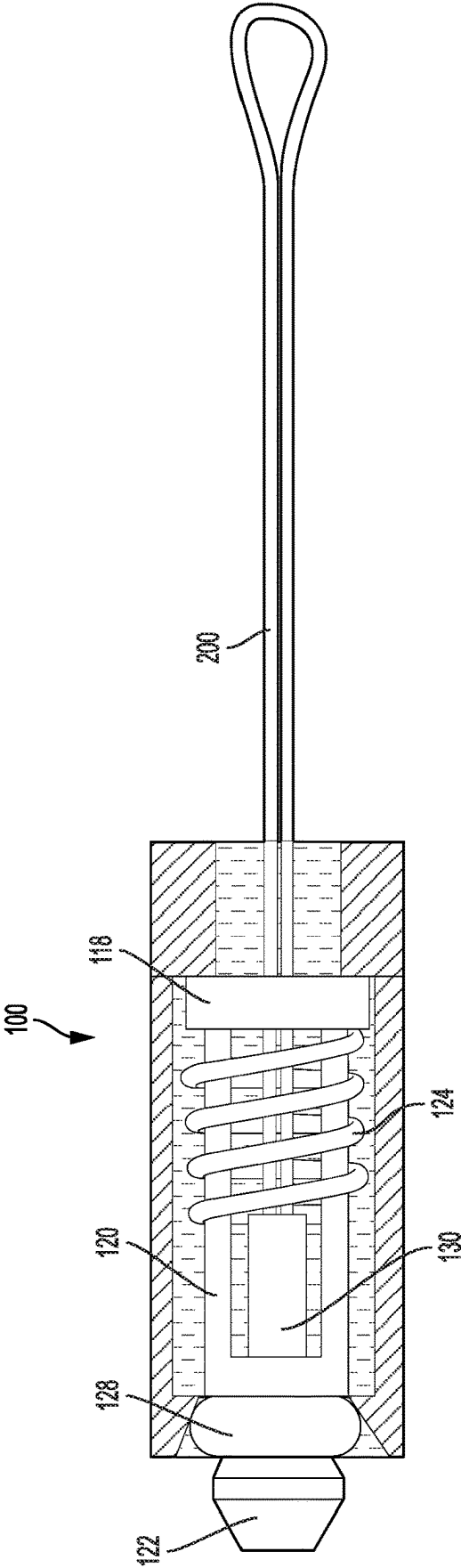


FIG. 24

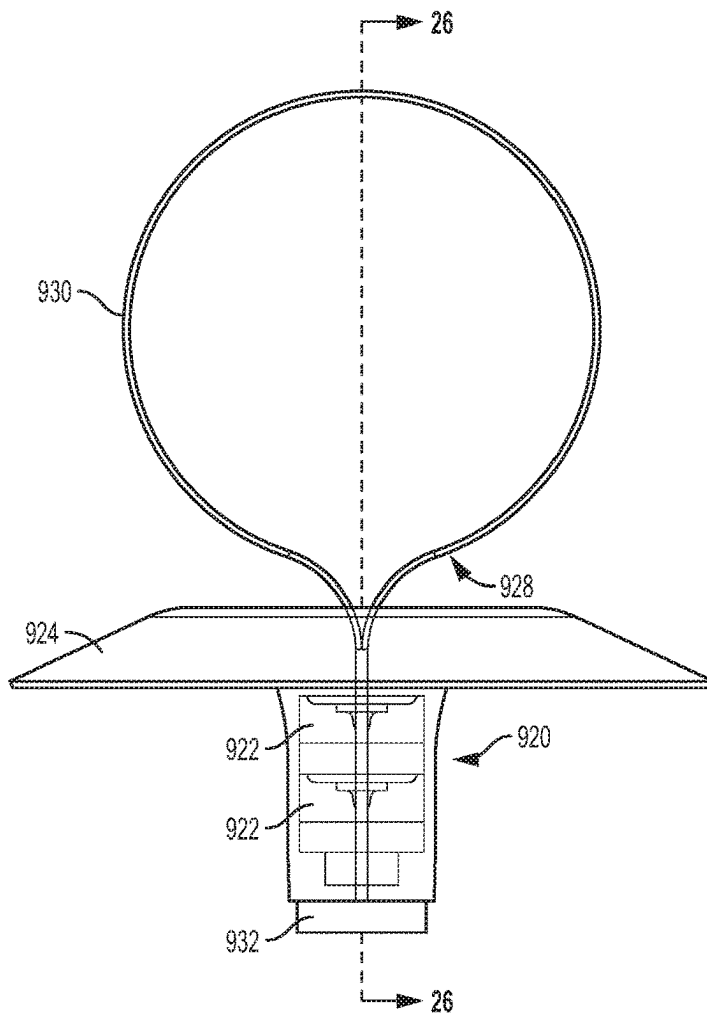


FIG. 25

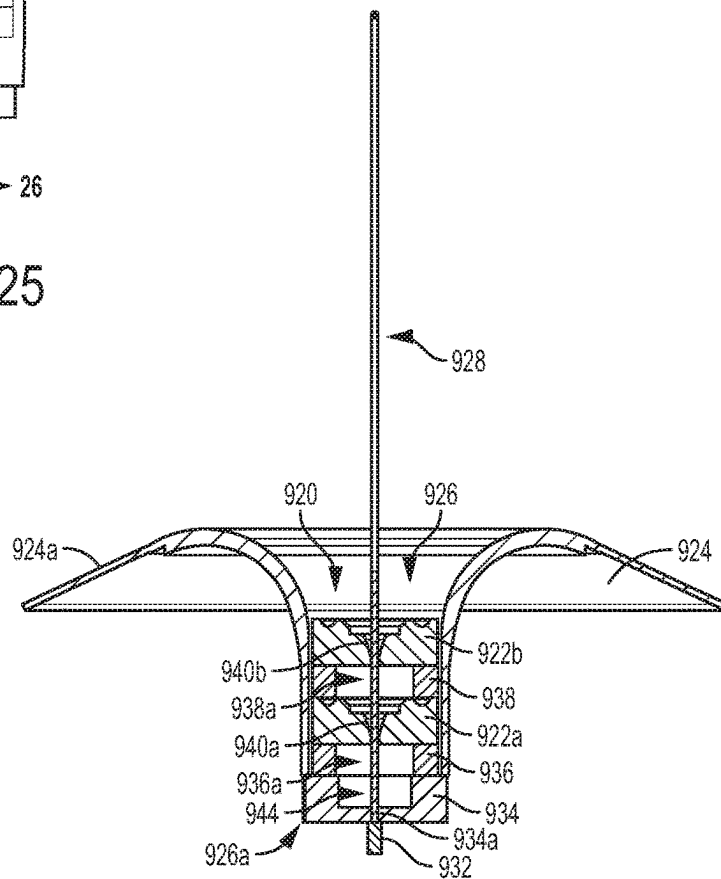


FIG. 26

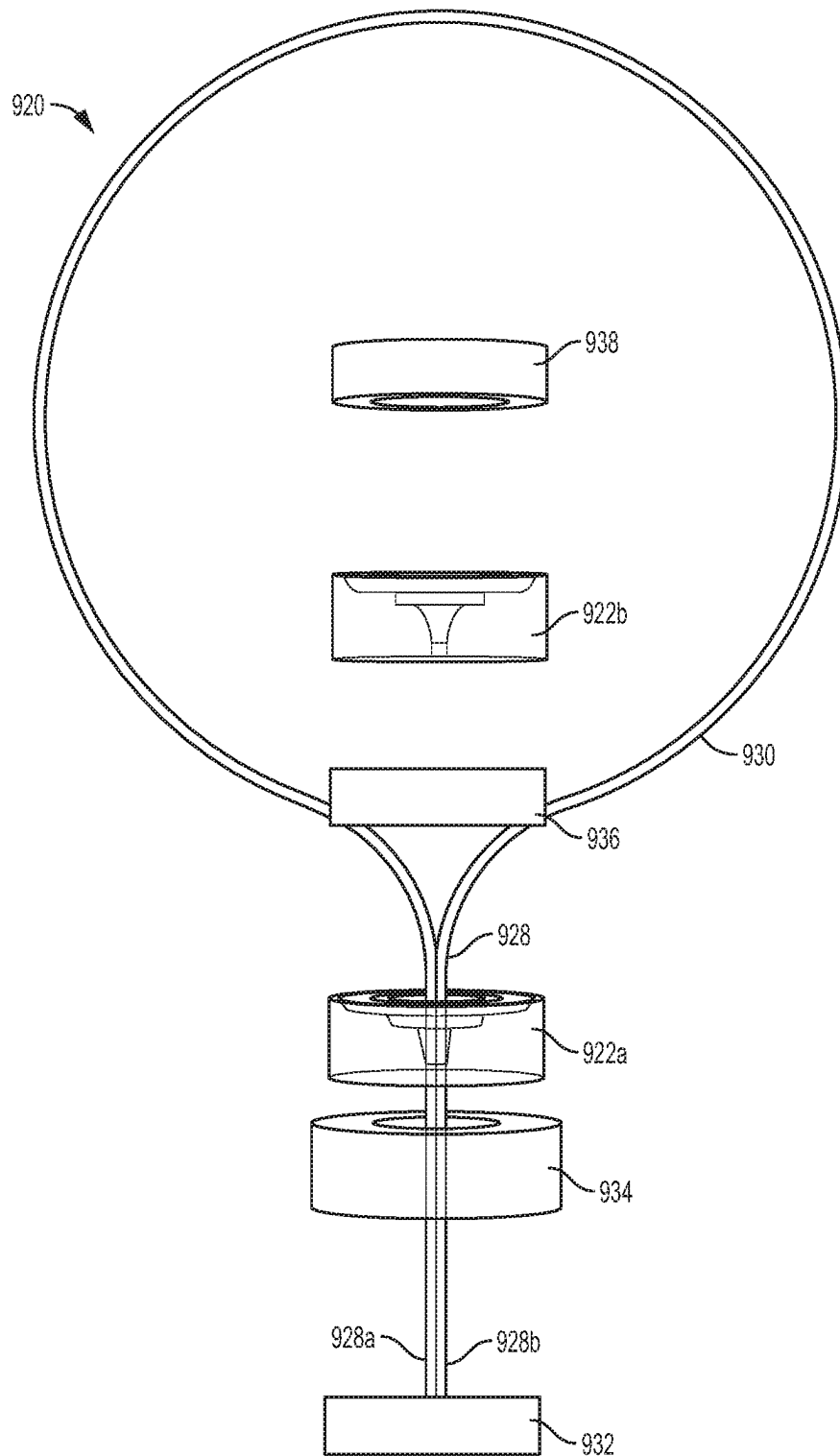


FIG. 27

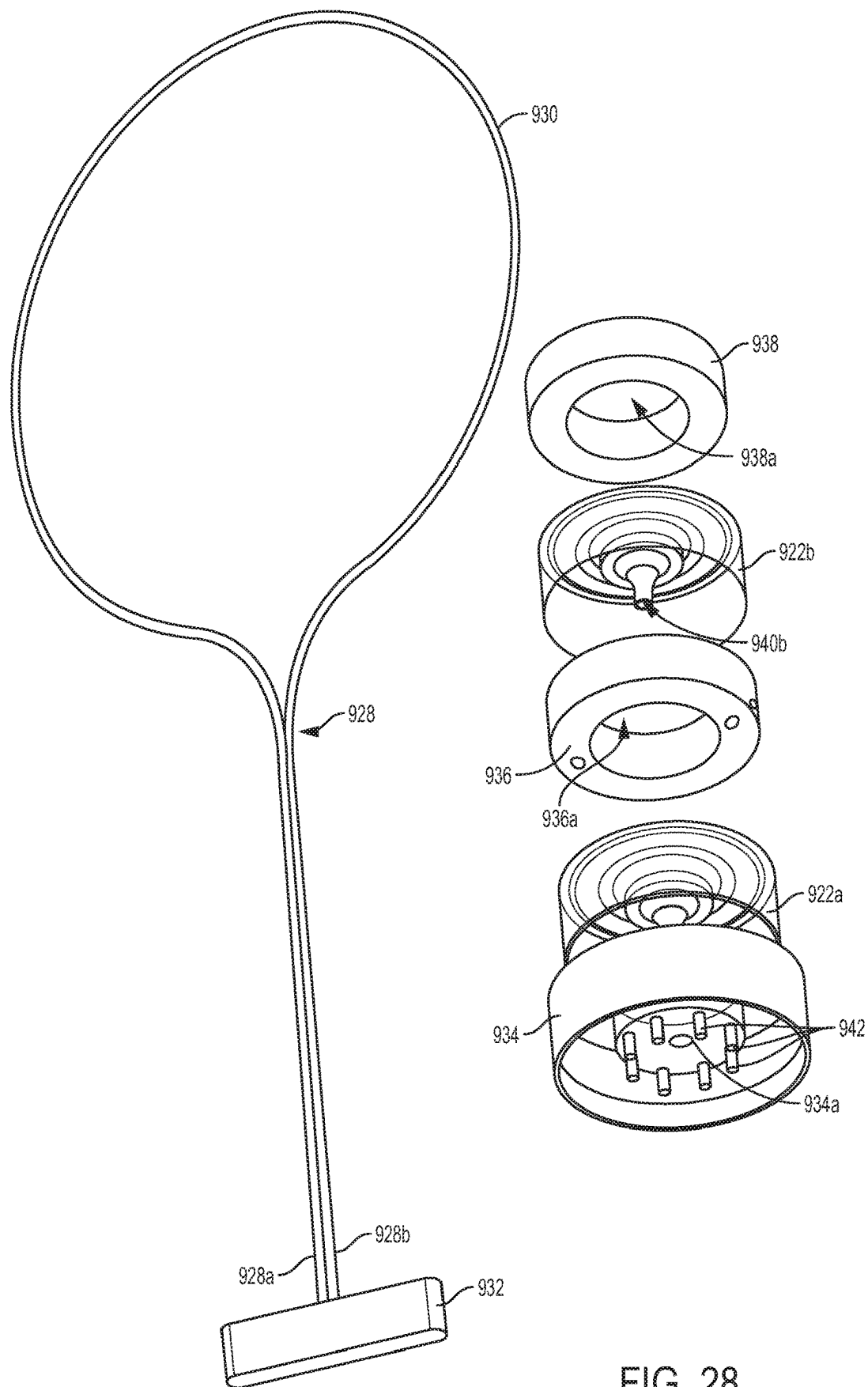


FIG. 28

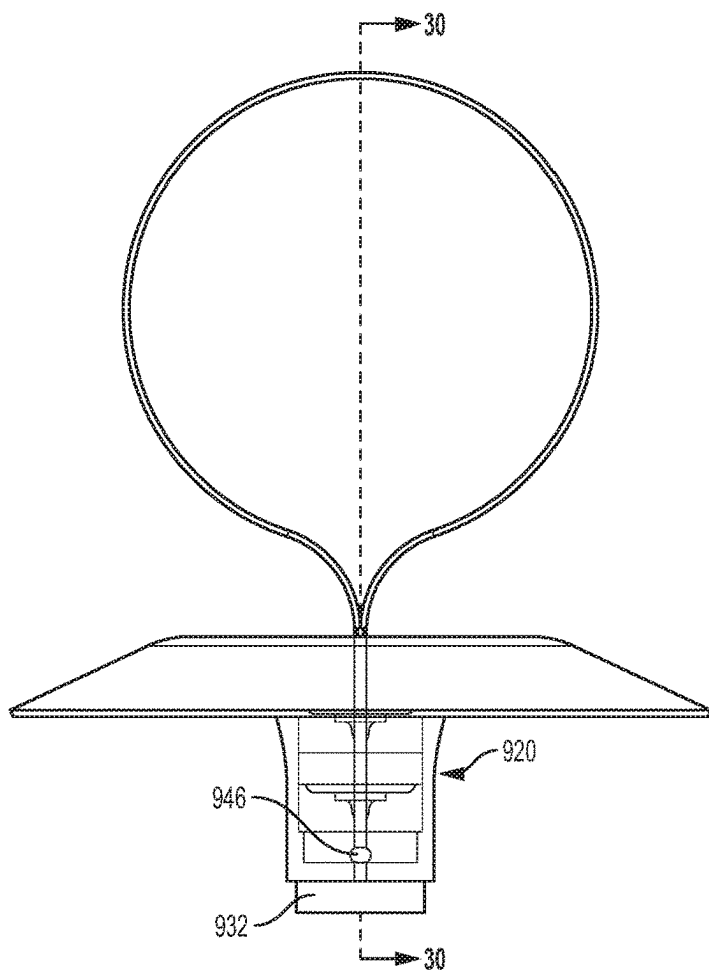


FIG. 29

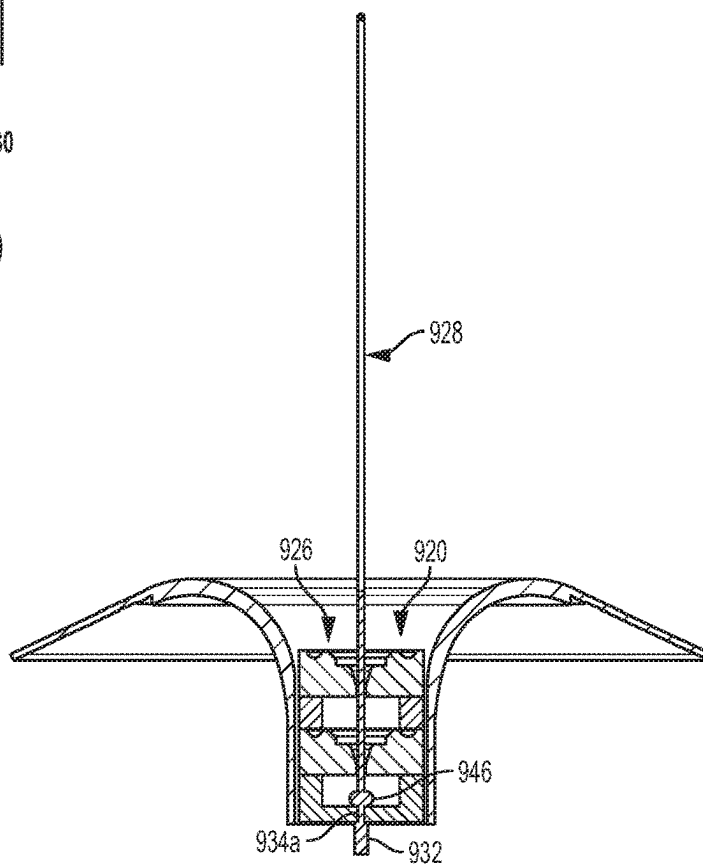


FIG. 30

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VALVE WITH DOCKING STATION FOR GASTROINTESTINAL BALLOON

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 62/784,106 filed Dec. 21, 2018, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD OF THE INVENTION

The present invention relates generally to implantable medical devices and systems that include a check valve, and methods of using implantable medical devices and systems with a check valve. Specifically, the present invention relates to devices, systems, and methods for a check valve and a gastrointestinal balloon including a check valve. The present invention also relates generally to a check valve with docking station. The check valve with docking station may be in a remote location or a difficult to access locations. The present invention relates to valves for access to a remote location, such as, a gastrointestinal balloon located in a patient. The present invention relates to a series of valves with a docking station.

BACKGROUND OF THE INVENTION

Morbid obesity remains an ever-growing problem in the U.S. Varying forms of gastric bypass surgery have developed and have improved over the last few decades. Recently, laparoscopic gastric bypass has emerged as a less invasive surgical option. However, bariatric surgery is fraught with morbidity of up to 20%, with a re-operation rate approaching 25% at 3-5 years post-operation. Bariatric surgery carries an operative mortality of 0.5%. Diet and pharmaceutical alternatives have not been very effective, with a high recidivism rate. Intragastic balloons in use may achieve weight loss and a drop in body mass index (BMI). However, it is desirable to be able to change the volume of the balloon while in the stomach of a patient, either to increase the balloon volume or decrease the balloon volume, as balloon effect wears off after several months and enlarging the balloon can rejuvenate the balloon effect, and some balloons may cause intolerance and require balloon volume decrease.

There is a need to be able to change the volume of a gastrointestinal balloon while the balloon is in the stomach of a patient. In order to access the balloon valve to adjust balloon volume while the balloon is still in the stomach, there are two options. One option is to bring the valve out of the stomach to the mouth of a patient, where balloon volume may be manually adjusted. For example, some existing gastrointestinal balloon systems use a stretchable inflation tube that may stretch from a stomach to the mouth of the patient, or about 2.5-10 times its length, to facilitate inflation or adjustment of volume of the balloon. The inflation tube is partly inside and partly outside of the balloon. U.S. Patent Application Publication No. 2006/0142731 and U.S. Pat. No. 8,403,952, which are incorporated herein by reference, describe a floating anchor that may be used, for example, with the gastrointestinal balloon. However, further improvements may be made to limit side effects from implantable devices including, but not limited to, tissue trauma, difficulty finding the tube, and difficulty grasping the tube, some of which may result of components of the devices that are located on the outside of the gastrointestinal balloon.

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Another option is to keep the valve stationary in the balloon and access the valve itself while the valve resides in the stomach, without removing any of the components from the stomach. However, direct access to the balloon valve while the valve is within the stomach via an endoscope is extremely difficult and may not be feasible because the positioning of the endoscope relative to the balloon valve requires pinpoint precision with head-on engagement. This is a technically difficult maneuver for an endoscope engaging a spherical balloon in the wet environment of the stomach. For example, the balloon may easily move within the stomach while, or as a result of, attempting to engage the valve.

Thus, there exists a need for improved gastrointestinal balloon devices, systems, and methods for accessing a balloon valve while it is in the stomach in a way that provides a high success rate for engaging with the valve to adjust the balloon volume.

BRIEF SUMMARY OF THE INVENTION

According to embodiments of the invention, a gastrointestinal balloon is immobilized within the stomach to improve the ease of engaging the valve. In particular, the balloon may be immobilized while at the same time allowing access to the valve at the correct angle by the inflation tool. A check valve is a type of valve that may be used for changing the volume of a gastrointestinal balloon. However, a standard check valve cannot be simultaneously immobilized and accessed with current check valve configurations and technology.

Thus, novel devices, systems, and methods are provided herein for a check valve used in inflatable medical devices, including but not limited to gastrointestinal balloons that may be simultaneously immobilized and accessed while in a body cavity. After immobilization of the balloon and aligning of the inflation tool to the balloon valve, the balloon may be inflated or deflated.

Additional features, advantages, and embodiments of the invention are set forth or apparent from consideration of the following detailed description and drawings. Moreover, it is to be understood that both the foregoing summary of the invention and the following detailed description are exemplary and intended to provide further explanation without limiting the scope of the invention as claimed.

According to an embodiment of the present disclosure, a check valve for use in a gastrointestinal balloon may include a housing having an entry, the entry configured to allow fluid flow into and out of the gastrointestinal balloon; a plunger configured to open and close the entry; and an attachment extending from a proximal end of the housing. The attachment is configured to be grasped by a tool to permit aligning of a catheter with a proximal end of the housing.

The check valve includes an open position and a closed position, wherein the check valve is configured to be moved to the open position in a stomach. The check valve is configured to be mobile in a stomach. The catheter is configured to impact the proximal end of the housing. The plunger further comprising a plunger head, a plunger arm, and a plunger tail. The plunger head includes an opening for the attachment to extend through and a seal for sealing the opening. The plunger arm is formed from a cylindrical body with two semi-circular arms extending between the cylindrical body and a distal end of the plunger head. The plunger tail is coupled to a distal end of the plunger arm. The plunger head, plunger arm, and plunger tail are integrally formed as a single unit. An o-ring seal is located between the plunger

arm and the plunger tail. A block is coupled to the housing and coupled to the attachment. A spring is coupled between the block and a proximal end of the housing. An opening in the proximal end of the housing is configured to removably receive a distal end of the catheter.

According to an embodiment of the present disclosure, a gastrointestinal balloon system, may include a gastrointestinal balloon; a valve located within the gastrointestinal balloon, the valve configured to allow fluid flow into and out of the gastrointestinal balloon; an attachment coupled to the valve, the attachment having a connector on a proximal end; and a shaft having a lumen, the lumen configured to receive a grasping tool. The grasping tool extends out a distal end of the shaft for grabbing the connector to align the shaft with the valve.

The shaft is a catheter and the catheter has a tapered distal tip. The valve separates an interior of the gastrointestinal balloon with an exterior of the gastrointestinal balloon. The gastrointestinal balloon is configured to be inflated or deflated through the valve. The valve includes a proximal opening, a plunger, and an o-ring to seal an entry into the gastrointestinal balloon. The valve is a check valve. The attachment is a string.

According to an embodiment of the present disclosure, a method of accessing a gastrointestinal balloon may include inserting a catheter through a channel of a gastroscope and into a stomach of a patient; inserting a grasping tool through a lumen of the catheter and extending the grasping tool through a distal end of the catheter; grasping an attachment with the grasping tool, wherein the attachment is coupled to a check valve; moving the check valve into alignment with the catheter; moving a plunger in the check valve such that the check valve is moved from a closed position to an open position; and inflating or deflating the gastrointestinal balloon through the check valve.

The method includes removably impacting the catheter into the check valve. The method includes impacting the catheter into the check valve comprises impacting a distal end of the catheter with a proximal opening of a housing of the check valve. The method includes moving the plunger in the check valve comprises distally moving the plunger in the check valve. The method includes releasing the attachment from the grasping tool and removing the grasping tool from the catheter prior to inflating or deflating the gastrointestinal balloon. The distal end of the catheter is tapered.

According to an embodiment of the present disclosure, a check valve system may include a check valve having a housing, the housing having a distal opening and a proximal opening, the distal opening configured to be sealed; an elongated member coupled to the housing, the elongated member having a proximal end with a connector; a tool configured to grasp the connector; and an impacting member configured to engage a proximal end of the housing. The tool is inserted through the impacting member to grasp the connector to align and immobilize the check valve with the impacting member. The tool is subsequently pulled to engage the impacting member into the proximal end of the housing to open the distal opening of the check valve. The check valve is configured for installation in a stomach, in a pipeline, in a plumbing line, or in a body cavity. A spring is configured to bias the check valve to a closed position.

According to an embodiment of the present disclosure, a two-way valve may include a body configured to allow two-way flow. The body may have a docking end configured to receive a tool for positioning the body in a first position; and a port end configured to be disposed in a cavity. The

valve has the first position and a second position. The docking end further includes an attachment to align the tool with the docking end.

The first position is an open configuration and the second position is a closed configuration, the valve biased to the second position. The two-way valve is a check valve. The attachment is a strap. The docking end comprises an opening in a proximal end of the body. The opening is configured to receive the tool for releasably impacting the plunger.

According to an embodiment, a valve for a gastrointestinal balloon, the valve comprising: a first valve formed of a self-sealing material; and an attachment extending from an interior of the gastrointestinal balloon through the first valve to an exterior of the gastrointestinal balloon.

According to an embodiment, the valve further comprising a second valve located in series with the first valve, the second valve formed of a self-sealing material, wherein the attachment extends through the second valve.

According to an embodiment, the valve further comprising a first spacer separating the first valve from the interior of the gastrointestinal balloon, the attachment extending through the first spacer.

According to an embodiment, the valve further comprising a second spacer located in series and between the first valve and the second valve.

According to an embodiment, wherein a first opening through the first valve and a second opening through the second valve are aligned.

According to an embodiment, wherein the first opening and the second opening each have a first state preventing fluid flow therethrough and a second state allowing passage of a tool.

According to an embodiment, wherein, in the first state, the first opening and the second opening are closed around the attachment.

According to an embodiment, wherein the self-sealing material is silicone.

According to an embodiment, wherein the first valve is formed as a cylindrical disc with an opening therethrough, the opening configured to receive the attachment.

According to an embodiment, further comprising an open position and a closed position, wherein the valve is configured to move to the open position in a stomach.

According to an embodiment, wherein the open position permits passage of a tool and the closed position prohibits flow of fluid.

According to an embodiment, further comprising a block configured to prevent removal of the attachment from the valve.

According to an embodiment, the attachment extends through a center of the first valve.

According to an embodiment, a gastrointestinal balloon system may include a self-sealing valve; an attachment extending through the self-sealing valve; and a tool configured to track over the attachment and pass through the self-sealing valve.

According to an embodiment, wherein the tool is configured to centrally track over the attachment such that a central axis of the tool and the attachment are substantially coaxial.

According to an embodiment, the self-sealing valve comprises a first self-sealing valve and a second self-sealing valve located in series.

According to an embodiment, the self-sealing valve has an open position and a closed position, wherein, in the open position, the tool and the attachment pass through an open-

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ing of the self-sealing valve, and wherein, in the closed position, only the attachment passes through the opening of the self-sealing valve.

According to an embodiment, a method of accessing a gastrointestinal balloon, the method comprising: grasping an attachment extending through a self-sealing valve; tracking a tool over the attachment and toward the self-sealing valve; pushing the tool through an opening in the self-sealing valve, still tracking over the attachment, thus opening the self-sealing valve; and inflating or deflating the gastrointestinal balloon through the tool.

According to an embodiment, wherein tracking the tool over the attachment comprises passing the attachment through a lumen of the tool.

According to an embodiment, wherein pushing the tool through an opening in the self-sealing valve comprises expanding the opening by entering a distal end of the tool into the opening.

According to an embodiment, wherein the self-sealing valve is a series of two self-sealing valves formed of silicone.

According to an embodiment, further comprising removing the tool from the self-sealing valve thus closing the self-sealing valve.

According to an embodiment, a system for accessing an interior of a gastrointestinal balloon may include a gastrointestinal balloon; two contiguous valves; and a string running through a center of each of the two contiguous valves and exiting the gastrointestinal balloon, wherein the string is configured to be grasped by a user to track a tool over the string and directly through the two contiguous valves and into the interior of the gastrointestinal balloon.

According to an embodiment, the two contiguous valves comprise a first valve located in series with a second valve, a first opening of the first valve and a second opening of the second valve being coaxially aligned.

According to an embodiment, the two contiguous valves abut and touch each other.

According to an embodiment, the system may include a spacer between the two contiguous valves.

According to an embodiment, the system may include a tool configured to centrally track over the string and pass through the two contiguous valves.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate preferred embodiments of the invention and together with the detailed description serve to explain the principles of the invention. In the drawings:

FIG. 1 shows a standard check valve attached to a balloon, with the valve in the closed position.

FIG. 2 shows the standard check valve of FIG. 1, with the valve in the open position.

FIG. 3 shows a check valve in a gastrointestinal balloon according to an embodiment of the present invention.

FIG. 4 shows a close-up view of the check valve of FIG. 3, with the valve in the closed position.

FIG. 5 shows a close-up view of the check valve of FIG. 3 from a perspective rotated 90° relative to the view shown in FIG. 4.

FIG. 6 shows a close-up of the check valve of FIG. 3 in an open position according to an embodiment of the present invention.

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FIG. 7 shows a catheter to be used in conjunction with the balloon and check valve and a step in a method of operating the check valve of FIG. 3, according to an embodiment of the present invention.

FIG. 8 shows a portion of the catheter, according to an embodiment of the present invention.

FIG. 9 shows a check valve in operation with a catheter and a grasping tool, according to an embodiment of the present invention.

FIG. 10A shows an exemplary grasping tool in a closed position, according to an embodiment of the present invention.

FIG. 10B shows an exemplary grasping tool in an open position, according to an embodiment of the present invention.

FIG. 11 shows a biluminal connector, according to an embodiment of the present invention.

FIG. 12 shows the biluminal connector of FIG. 11 with a coupling sleeve, according to an embodiment of the present disclosure.

FIG. 13 shows the biluminal connector and coupling sleeve of FIG. 12 coupled to a gastroscope, according to an embodiment of the present disclosure.

FIGS. 14A-14E show cross-sectional views of the biluminal connector of FIG. 11 taken along various axes of FIG. 11, according to an embodiment of the present disclosure.

FIG. 15 shows an alternative method for grasping the string, according to an embodiment of the present disclosure.

FIG. 16 shows a check valve in a gastrointestinal balloon according to another embodiment of the present invention.

FIG. 17 shows a close-up view of the check valve of FIG. 16, with the valve in the closed position.

FIG. 18 shows a close-up view of the check valve of FIG. 16 from a perspective rotated 90° relative to the view shown in FIG. 17.

FIG. 19 shows a close-up of the check valve of FIG. 16 in an open position according to an embodiment of the present invention.

FIG. 20 shows a catheter to be used in conjunction with the balloon and check valve of FIG. 16 and a step in a method of operating the check valve, according to an embodiment of the present invention.

FIG. 21 shows a check valve in operation with a catheter and a grasping tool, according to an embodiment of the present invention.

FIG. 22 shows a check valve in an open position with fluid flowing therethrough, according to an embodiment of the present invention.

FIG. 23 shows the check valve of FIG. 22 in the open position, rotated 90 degrees from the view of FIG. 22, according to an embodiment of the present invention.

FIG. 24 shows the check valve of FIG. 22 in the closed position, with fluid isolated on either side of the valve, according to an embodiment of the invention.

FIG. 25 shows a valve attached to a balloon, according to an embodiment of the present invention.

FIG. 26 shows a cross-sectional view of the valve of FIG. 25 taken along the section line 26-26.

FIG. 27 shows an exploded view of the valve of FIG. 25.

FIG. 28 shows an exploded view of the valve of FIG. 25.

FIG. 29 shows a valve attached to a balloon, according to an embodiment of the present invention.

FIG. 30 shows a cross-sectional view of the valve of FIG. 29 taken along the section line 29-29.

DETAILED DESCRIPTION OF THE INVENTION

Devices, systems and methods are described herein with reference to the figures and in the context of medical devices, in particular a gastric balloon. In one aspect, a check valve with docking station can be provided to improve the ability to control the check valve for purposes of adjusting a gastric balloon (such as inflating and deflating the balloon). The check valve with docking station can improve the user's ability to change the volume of a gastrointestinal balloon while the balloon is in the stomach of a patient. The check valve with docking station can be used in other environments in order to access and adjust the valve in hard to reach areas where the devices may or may not be stabilized. The check valve docking features can be implemented in other medical devices as well as non-medical devices and used as described herein in accordance with the principles of the invention. The check valve may include a plunger, string, spring, and valve housing. The string may be coupled to the housing such that when grasped and pulled, it aligns an opening of the valve housing with an impacting member for operating the opening and closing of the valve. Thus, the valve may be controlled from a remote location to allow fluid to flow into or out of the valve, as will be appreciated from the disclosure to follow.

A standard check valve **10**, as shown in FIGS. **1** and **2** (in the closed and opened positions, respectively), cannot be simultaneously immobilized and accessed with current check valve configurations and technology. FIGS. **1** and **2** depict a check valve **10** which separates the exterior **12** of the balloon from the interior or cavity **14** of the balloon. The check valve **10** may include a housing **16**, plunger head **18**, plunger arm **20**, and a plunger tail **22**. In FIG. **1**, a spring **24**, at a full extension length L_E , keeps the check valve **10** in a closed position by pushing plunger head **18** away (to the left in the view shown in FIG. **1**) from the mouth or entry **26** into the balloon cavity **14**. The plunger head **18** in turn pulls the plunger arm **20** and the plunger tail **22** to the left and closes the entry **26** into the balloon cavity **14** by pulling an o-ring **28** to the left. In FIG. **2**, The plunger head **18** is pushed toward (to the right in the view shown in FIG. **2**) the entry **26** which contracts the spring **24** to a length L_C , thereby allowing the plunger head **18** to move to the right (FIG. **2**) towards the balloon cavity entry **26**. The plunger head **18** then pushes the plunger arm **20** and the plunger tail **22** to the right which pushes the o-ring **28** to the right and opens for the bi-directional flow of fluid (arrows **F**) into and out of the balloon cavity **14**.

Engaging the check valve **10** in the balloon while the check valve **10** sits in the body cavity (e.g. the stomach) requires some method of immobilization and proper alignment of a tool on the check valve **10** of the balloon, followed by applying force to push the spring **24** towards the balloon cavity **14** (thereby opening flow **F** to the balloon cavity **14**). Therefore, the present invention provides a novel modification to check valve technology that may accomplish these goals. The novel check valve has utility not only in the intragastric adjustable balloon, but also in other inflatable medical devices in any body cavity or check valves for industrial and non-medical use, such as in plumbing. Therefore, references to "gastrointestinal balloons" or "stomach" are for example only, and are not intended to limit the scope of the invention.

FIG. **3** shows a valve **100** according to an embodiment of the invention. The valve **100** may be a check valve, two-way valve, two directional flow valve, or other valve having an

opening configured to be selectively opened and closed in accordance with the principle of the invention. For ease of description, the valve **100** is referred to herein as a check valve **100**. The check valve **100** may include a docking station to allow for alignment and engagement of an impacting member to open and close the check valve **100**. The check valve **100** is shown within a balloon **300** or integral with a sidewall **302** of the balloon **300**. The balloon **300** may be an adjustable balloon located in a patient's stomach. Alternatively, the check valve **100** may be located in a flow line of a plumbing or other hydraulic system. The balloon **300** may have an interior cavity **114** separated from an exterior space **112** (e.g. the stomach environment). The check valve **100** may separate the interior cavity **114** of the balloon **300** from the exterior space **112** of the balloon **300**. The check valve **100** may be provided with an attachment **200**, such as an elongated member **200**, such as a string **200** that may have a connector **202**. The connector **202** may be an "easy grasp" loop **202** that exits into the exterior space **112** of the balloon **300**. The string **200** with easy grasp loop **202** may be formed of nylon. The attachment **200** may be an elongated member, a bungee, a stretchy member, a string, a cable, a cord, other elongated device, or anything in accordance with the principles of invention capable of coupling to the check valve **100** and being grasped by a tool. For ease of description, the attachment **200** is described as a string **200** and the connector **202** is described as a loop **202**. The string **200** may be a single string **200** having opposing distal ends, each coupled to a block **130** (FIG. **4**) of the housing of the check valve **100** with the easy grasp loop **202** formed therebetween.

Moving (e.g. by pulling or pushing the impacting member and/or tool) the loop **202** into an impacting member **400** (FIG. **7**), such as a catheter, may immobilize the balloon **300** and may guide the inflation/pushing catheter (not depicted) to the target (i.e., the check valve **100**), as will be described to follow. The string **200** may have a proximal end having a loop (such as easy grasp loop **202**) configured to be grasped by a tool. The easy grasp loop **202** may be pulled into a catheter by a grasping tool dispatched through an endoscope, as will be described in more detail. As may be appreciated, the balloon **300** may be located in a patient's stomach. The balloon **300** (including check valve **100**) may be floating or otherwise mobile within the stomach as the balloon is not secured to any wall therein. As previously mentioned, inflation and deflation of the balloon **300** may be difficult due to the mobile nature of the balloon **300**. In conventional applications, the balloon **300** is removed from the stomach. With the present check valve **100**, attachment **200**, and docking station, the balloon **300** is capable of being inflated or deflated while the balloon **300** remains in the patient's stomach. That is, as will be described below, the check valve **100** may be immobilized and aligned with a tool, the tool configured to open the valve and allow flow into or out of the balloon **300** as necessary. The fluid allowed to flow through the check valve **100** may be liquid, air, inflation fluid, or other fluid in accordance with principles of the invention.

FIG. **4** depicts a close-up view of the check valve **100** with the balloon **300** omitted for clarity. The check valve **100** in FIG. **4** is depicted in a closed position. The check valve **100** may be biased in the closed position by a spring **124**. In this manner, once inflated or deflated to the appropriate level, the check valve **100** may retain the balloon **300** (not depicted) in the desired state. The check valve **100** may include a housing **116**, a plunger head **118**, a plunger arm **120**, a plunger tail **122**, the spring **124**, an o-ring **128**, and a block

130. The housing 116 may be formed as a single piece or may be formed as a first housing portion 116a and a second housing portion 116b coupled together. The entry 126 may be formed integral with an interior surface of the housing 116 or may be a separate piece secured (e.g. with threads or fasteners) to an interior surface of the housing 116. The block 130 may be built into, integrally formed with, or otherwise secured to the housing 116. Thus, the block 130 may be a stationary, non-moving component against which the spring 124 may be compressed. Alternatively, the block 130 may be permitted to move a small distance until engagement with an interior surface of the housing 116. As may be appreciated, the spring 124 may be compressed between the surface 130a of the block 130 and a distal surface of the plunger head 118.

In the check valve 100, the plunger head 118, plunger arm 120, and plunger tail 122 may all be connected as one unit, similar to the check valve 10 of FIG. 1. Alternatively, the plunger head 118, plunger arm 120, and plunger tail 122 may be formed integrally. The plunger arm 120 may be formed as a solid piece as is the plunger arm 20 of FIG. 1. Alternatively, the plunger arm 120 may be split in the middle, leaving space for other parts, such as the string 200, to be connected to the block 130 within the housing 116 of the check valve 100, as will be described in relation to FIG. 5.

With continued reference to FIG. 4, in the closed position, the spring 124 may be extended to a length L_E pushing the plunger head 118 to the left in FIG. 4. In the closed position, the plunger tail 122 may hold the o-ring 128 in a sealing position with the entry 126 to the interior cavity (114 in FIG. 3) of the balloon 300 (FIG. 3). The closed position of FIG. 4 may be the normal position of the check valve 100. That is, the spring 124 may bias check valve 100 to a normally closed position. It may be appreciated that the plunger 118, 120, 122 may slide with respect to the block 130. That is, the block 130 may extend through an interior of the plunger arm 120 (FIG. 5) and the plunger may move with respect to the block 130 to compress the spring 124 against the block 130 and extend the o-ring 128 off of the entry 126 allowing flow therethrough.

Referring now to FIG. 5, a close-up view of the check valve 100 of FIG. 4 is shown, rotated 90° from the perspective of FIG. 4. The plunger head 118, plunger arm 120, and plunger tail 122 of the check valve 100 may be formed as a single, integral piece or may be formed as separate parts coupled or secured together. The plunger head 118 may be generally cylindrical with a hole (not shown) through which the string 200 may pass. The plunger head 118 may be disc shaped. The hole may be located in the center of the plunger head 118 or may be located anywhere through plunger head 118 to facilitate connection of the string 200 to the block 130. The plunger head 118 may further include a silicone seal 134 for sealing the hole and the string 200 within. The silicone seal may prevent fluid from passing through the hole having the string 200. The plunger head 118 may include a cut out or groove 136 for coupling to the plunger arm 120. Alternatively, the plunger head 118 and the plunger arm 120 may be formed as an integral piece.

With continued reference to FIG. 5, the plunger arm 120 may be cylindrical. The plunger arm 120 may be split in the middle such that a cavity 132 is located between opposing sides 120a and 120b of the plunger arm 120. At the proximal end, the plunger arm sides 120a, 120b may be coupled or integral with the plunger head 118. At the distal end, the plunger arm 120 main (cylindrical) body may be coupled to or integral with a neck (not shown) that is in turn coupled or

integral with the plunger tail 122. The neck may have a reduced diameter from the plunger arm 120 and/or plunger tail 122 such that the o-ring 128 may be fitted around the neck. The string 200 may have two ends 200a, 200b which extend through the plunger head 118 and through the cavity 132 of the plunger arm 120. The two ends 200a and 200b may then be coupled, secured, or attached to the block 130. Alternatively, the string 200 may be one or multiple strings with knots, loops, or components that may allow for grasping by a tool. As previously described, the plunger, including plunger arm 120, may be allowed to move with respect to the block 130 and the housing 116.

Referring now to FIG. 6, the check valve 100 is depicted in an open position, with the balloon 300 (FIG. 3) removed for clarity. In the open position, the plunger head 118 may be pushed to the right in FIG. 6 within the housing 116. Pushing or moving the plunger head 118 may shorten the spring 124 to the contracted length L_C . As may be appreciated, the spring 124 may be secured between a proximal surface 118a of the plunger head 118 and a distal surface 130a of the block 130. In this manner, the block 130 may operate as a stopper beyond which the spring 124 cannot pass. As the plunger head 118 is moved or pushed to the right, plunger arms 120 are correspondingly moved or pushed to the right due to the integral or coupled nature of the components. The plunger arms 120 move the neck (not depicted) and plunger tail 122 into the interior cavity 114 (FIG. 3) of the balloon and out of the housing 116. As the plunger tail 122 is moved out of the housing 116, the o-ring 128 is moved away from the entry 126, thus unseating the o-ring 128 from the entry 126 and allowing flow into and out of the interior cavity 114 of the balloon 300.

FIG. 7 shows an embodiment of an impacting member as a catheter 400, which may, for example, have a 2-3.5 mm outer diameter with a wall thickness of 0.1 to 0.6 mm. The catheter 400 may be made of a semi-flexible material that allows the catheter to assume curves up to 90 degrees, yet at the same time maintain rigidity. Materials such as Teflon, low friction synthetic polymers, polymers coated with materials that produce a non-friction surface such as parylene or Teflon, and other material types may be employed. The catheter 400 may have a lumen with a diameter of 0.8 to 3.3 mm. The catheter 400 may be used in conjunction with the check valve 100. The catheter 400 may be made of any suitable biocompatible material or material used for medical devices. The catheter 400 may have a tip 402, but is not limited to the shape shown in FIG. 7. The tip 402 may be tapered, a luer, cone shaped, straight, or rounded tip. The tip 402 may facilitate entry of the catheter 400 into an opening 116c of the housing portion 116b. The opening 116c may be a proximal opening of the housing 116 and the entry 126 may be a distal opening of the housing 116. The catheter 400 may have a central lumen 404 through which a grasping tool 500 may extend. The grasping tool 500 may grab or secure the easy grasp loop 202 (FIG. 3) of the string 200, as will be described to follow. The tip 402 may encounter the plunger head 118 during operation of the check valve 100 to move the plunger head 118 and thus open the check valve 100. As may be appreciated, the proximal end of the housing portion 116b, opening 116c, the plunger head 118, and the string 200 may be considered a docking station for the check valve 100. That is, the catheter 400 is aligned with and impacted with these components such that alignment and operation of the check valve 100 are permitted. In this regard, the components act as a docking station for the catheter 400 and tool 500.

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Referring to FIG. 8, a proximal part 406 of the catheter 400 may be Y-shaped with a homeostatic valve 408 at each arm that may be tightened or loosened to regulate flow and thereby prevent backflow outside of the catheter 400. The proximal part 406 may include an inflation and/or deflation site 410 for inflating and/or deflating the balloon 300. The proximal part 406 may have a handle with mechanical control that closes and opens the hook 502 (FIG. 10) of grasping tool 500, or a handle that closes and opens the grasper of any type of grasping device.

Referring to FIG. 9, during operation, an endoscope is extended into a patient's stomach. A catheter 400 with a grasping tool 500 are inserted therethrough. The grasping tool 500, such as grasping forceps used in endoscopic procedures may be inserted into the lumen 404 of the catheter 400 through the homeostatic valve 408 (FIG. 8). During the procedure, the balloon, check valve, catheter, and grasping tool may all be located within the patient's stomach. The forceps or similar grasping tool 500 may grasp the string 200 by the easy grasp loop 202 and pull the check valve 100 toward the catheter 400. At the same time, the catheter 400 may be advanced toward the check valve 100. This may easily guide the grasping tool 500 to the target, the opening 116c, of the check valve 100. The catheter 400 then enters the opening 116c of the housing 116 of the check valve 100. Such alignment of the catheter 400 with the check valve 100 allows for immobilization of the check valve 100.

While fixating the catheter 400 and gastroscope with one hand, the grasping tool 500 (now grasping string 200) within the catheter 400 is pulled back with the other hand. This motion causes the tip 402 to releasably engage the plunger head 118 of the check valve 100. Continued pulling of the string 200 pushes the catheter tip 402 into releasably impacting the housing 116 and thereby causing the plunger head 118 to move to the right (as shown in the figures) or towards the interior cavity 114 of the balloon 300. This movement causes the plunger arm 120 and the plunger tail 122 to move to the right which thereby unseats the o-ring 128 and opens the check valve 100. At the same time, the tip 402 of the catheter 400 impacts into the housing 116. Alternatively, the catheter 400 may be straight, and the opening 116c of the housing may be cone shaped resulting in the same impaction effect. The catheter 400 may impact into the housing 116, which maintains the check valve 100 in an open position. The distal movement of the plunger head 118, the plunger arm 120, and the plunger tail 122, the check valve 100 opens to bi-directional flow by separating the o-ring 128 from contacting the entry 126. Once the catheter 400 is impacted and the check valve 100 is open, the string 200 may be released by the grasping tool 500 and the grasping tool 500 may be pulled out of the catheter 400. This may open more space in the lumen 404 and allow enhanced flow of fluid. Alternatively, the grasping tool 500 may be left in place and flow may continue at a slower pace.

After inflation or deflation is completed, the catheter 400 may be removed from the docking station or entry to the check valve 100. The tool 500, catheter 400, and endoscope may be removed from the patient's stomach and from the patient.

The force of the catheter 400 on the check valve 100 to impact and disengage the check valve 100 may be between 0.8 kg and 2 kg. The forces to contract and relax the spring 124 may be less than the forces that impact the valve by 25% to 40%. If the spring forces are greater than the impaction forces then the spring 124 may disengage the catheter 400 from the check valve. 100

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With continued reference to FIGS. 3-9, and described in another manner, during operation, the grasping tool 500 exits the endoscope and grasps the easy grasp loop 202 of the string 200. The grasping tool 500 may be pulled while the tip 402 of the catheter 400 has entered into the check valve 100 of the balloon 300. The catheter 400 may enter with ease, until the tip 402 broadens and impacts the plunger head 118 inside the housing 116 of the check valve 100. The grasping tool 500 may be pulled hard to impact the catheter 400 into the check valve 100. If the catheter 400 is not impacted into the check valve 100, the check valve 100 may close prematurely. As the catheter 400 impacts inside the housing 116, the catheter 400 pushes the plunger head 118 toward the entry 126 (compressing the spring 124). This in turn pushes the plunger arm 120 and the plunger tail 122 forward, unseating the o-ring 128 from the entry 126, and opens the check valve 100 to bi-directional flow. At this point the grasping tool 500 may release the string 200 (since the cone-tipped catheter 400 is impacted inside the housing 116). The grasping tool 500 may then be pulled out of the catheter 400. At this time, fluid may be admitted from the inflation site 410 (or removed through the deflation site 410), though the lumen 404 of the catheter 400, through the entry 126 of the check valve 100 and into the interior cavity 114 of the balloon 300. With reference to FIG. 6, the flow may enter the opening 116c of the housing 116, flow around the plunger head 118, around and through the plunger arms 120, through the entry 126, and past the o-ring 128 and plunger tail 122. It may be appreciated that during deflation, flow may follow the same path in the reverse direction. Once inflation/deflation is completed, the catheter 400 is pulled out of the check valve 100 and the spring 124 expands to its preferred position, which brings the plunger head 118, plunger arm 120, and plunger tail 122 back to the closed position (FIG. 5) with the o-ring 128 sealed in the entry 126 and flow of fluid prevented.

Although the catheter 400 is described as impacting the check valve 100, the catheter may be forced into the valve in other manners. For example, the catheter 400 may be secured to the check valve opening 116c with a turning motion. That is, the catheter 400 may have a tip 402 which may be screwed or threaded into the valve opening 116c. One may appreciate that the inner surface of opening 116c and the outer surface of the catheter tip 402 may have complementary threads or other surfaces to allow the rotational and longitudinal movement.

FIGS. 10A and 10B depict the grasping tool 500 in an open (FIG. 10A) and closed (FIG. 10B) position. In the open position, the grasping tool 500 may have a hinged arm or hook 502 (or other connection type) that may align with the easy grasp loop 202 (FIG. 3) of the string 200. Once properly aligned, the hinged arm or hook 502 may be actuated to close thus enclosing the easy grasp loop 202 and securing the grasping tool 500 to the check valve 100 via the string 200. Once grasped, the grasping tool 500 may be pulled to pull the string 200 and thus the check valve 100 and balloon 300 into alignment with the catheter 400. After proper alignment, the aforementioned impacting and opening of the check valve 100 may be performed. The grasping tool 500 may then be opened to release the check valve 100 (after impaction and/or opening). The grasping tool 500 may have a curved end that is hinged and can be opened to a straight position. In the closed position (FIG. 10B), the length of the hinged arm or hook 502 is within 0.5 to 2 mm of the inner diameter of the catheter. The hinged arm or hook 502 may be moved into a position that can grasp (FIG. 10A) or to a position that does not grasp (FIG. 10B). The grasping

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tool **500** may alternatively be other types of grasping tools, such as a rat-tooth grasper or hook grasper.

Thus, as may be appreciated from the foregoing disclosure, during a procedure to inflate and/or deflate the balloon, a catheter may be inserted into the channel of a standard flexible gastroscope, as practiced routinely by those skilled in the art of gastrointestinal endoscopy. As previously mentioned, the check valve and method of aligning and opening the check valve may be utilized in other body cavities or for industrial, plumbing, pipelines, wellbores, or other non-medical uses. For example, a check valve may be located in a pipeline. The pipeline may be inaccessible from the outer surface (e.g. a buried pipeline). The check valve may be biased to a normally closed position (as previously described) or biased to a normally open position. When it is desired to open or close the check valve, a grasping tool, similar to the grasping tool **500** but sized and dimensioned to fit in a pipeline instead of a catheter, may be extended into the pipeline inside another device (e.g. inside a tubular snake). Once inside the pipeline, the grasping tool may grasp a loop on a string coupled to the pipeline and align an impacting device (similar to the impacting catheter **400**) with an opening of the check valve. The grasping tool may pull the impacting device into alignment and impact the check valve in a manner similar to that previously described to effectuate opening and/or closing of the check valve.

In an alternative embodiment shown in FIGS. **11-14**, the flow of inflation/deflation fluid may be separated from the lumen having the grasping tool **500**. There may be a need to separate the grasping tool **500** from the flow of inflation fluid for fear of gastric fluid contamination. This may be accomplished using a biluminal connector **600**. In this embodiment, the catheter **400** may be replaced with the biluminal connector **600** having a proximal end **602** which sits in a distal end of a channel of a gastroscope **800** (FIG. **13**). The biluminal connector **600** may have a distal end **604** which impacts the check valve **100** (not depicted). The distal end **604** may have a tip **606** for guiding the tool into alignment into the check valve **100**. The tip **606** may be the same as or similar to the tip **402** of the catheter **400**. The biluminal connector **600** may further have an opening **608** in the side wall of the biluminal connector **600**. The opening **608** may allow entry and/or exiting of the inflation and/or deflation fluid.

The immobilization of the balloon with the grasping of the string can be performed with the grasping tool, as previously described. The grasping tool may be alone in the gastroscope channel and the inflation may be performed in a second channel, separated from the gastroscope channel. The biluminal connector **600** may have two lumens, as best depicted in FIGS. **14B** and **14C**. A first, central lumen **612** may extend from the proximal end **602** (see FIG. **14A**) to the distal end **604** (see FIG. **14E**). Once coupled to the gastroscope, the first lumen **612** may allow entry of the grasping tool **500** through the gastroscope channel and through the first lumen **612** to allow for grasping, alignment, and impaction of the check valve **100** as previously described. The first lumen **612** may be wholly separated from a second lumen **614** (FIGS. **14B**, **14C**). The second lumen **614** may be an annular lumen extending circumferentially around the outer surface of the first lumen **612** such that the first lumen **612** and the second lumen **614** are concentric. The second lumen **614** may be located distally to the gastroscope **800**. The second lumen **614** may be open distal to the gastroscope channel and allow entry from an outer side channel parallel to the gastroscope **800**. The entry may be located through opening **608** such that a flexible tube may be coupled to

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opening **608** to allow for bi-directional flow (inflation and/or deflation) through the flexible tube, second lumen **614**, and check valve **100** (not depicted). The external channel provided in the flexible tube may run parallel to the gastroscope **800** and then may turn in to the distal end of the gastroscope **800** and enter the second lumen of the biluminal connector **600**.

Referring to FIG. **12**, the biluminal connector **600** may have a silicone sleeve **700** to secure the biluminal connector **600** to the gastroscope **800**. The sleeve **700** may have an opening **708** which aligns with the opening **608** of the biluminal connector **600**. The biluminal connector **600** may be stabilized on the distal end of the gastroscope **800** (FIG. **13**) with the sleeve **700**. The biluminal connector **600** may further have a plug **610**. The plug **610** may have a rounded or atraumatic distal tip that may be placed into the distal opening of the biluminal connector **600**. The plug **610** may be pushed out by the grasping tool at any time. The plug **610** may be silicone or other flexible rubbery material or can be made of a material that melts at body temperature such as those used with medication capsules.

Referring to FIG. **13**, the sleeve **700** may fit snugly around the distal end of the gastroscope **800**. The inflation tube may be external to the gastroscope and enter through the opening **708**. The biluminal connector **600** may have the proximal end **602** sitting within the distal channel of the gastroscope **800**. The distal tip **606** may enter the check valve **100** (FIG. **3**) and move the plunger head, the plunger arm, and the plunger tail distally. Near the distal end of the biluminal connector **600**, the body widens with its tapering profile to impact within the check valve **100**. As may be appreciated, the biluminal connector may prevent contamination of inflation fluid with gastric fluid and may allow the gastroscope channel to utilize its suctioning function during the procedure.

FIGS. **14A-14D** depict cross-sectional views of the biluminal connector **600** taken along the lines A, B, C, and D of FIG. **1**. FIG. **14E** depicts a cross-sectional view of the biluminal connector **600** at the distal end. FIG. **14A** shows the first lumen **612** which sits within the gastroscope channel and allows entry of the grasping tool. FIG. **14B** shows the opening **608** through which the external inflation channel enters into the second lumen **614** which is an outer concentric lumen to the first lumen **612**. FIG. **14C** shows the concentric biluminal with supporting walls **616**. FIG. **14B** shows the first lumen **612** through which the tool travels and grasps the string **200** (FIG. **3**).

The biluminal connector **600** may be made from silicone or a firmer material, such as Teflon or other synthetic material. The biluminal connector **600** may have a wall thickness of 0.2 to 5 mm with a central lumen of 0.8 to 2.20 mm that accommodates a grasping tool with an external diameter of 0.7 to 3.0 mm. The biluminal connector **600** may be 6-15 mm long and 3-12 mm wide.

Referring to FIG. **15**, an alternative approach to accessing the balloon valve within the stomach is to separate the immobilization of the balloon from the accession of the valve. One method of immobilization is to place handles on the balloon (not depicted) that are grasped by the scope. Following this immobilization step, the valve (check valve, duck valve, or any type of valve) is accessed with a catheter that enters from the lumen of the endoscope into the valve and opens the valve, as previously described. Upon withdrawal of the catheter the valve closes. Following closure of the valve, the immobilization of the balloon is discontinued. As an example of the handles on the balloon one could place two loops of string (nylon for example) at the 3 o'clock and

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9 o'clock positions of the valve housing. The two loops may be grasped by two grasping forceps **900** that exit at the 3 o'clock and 9 o'clock positions in a sleeve **902** that surrounds the endoscope **904**. The two forceps **900** grasp the two loops (not depicted) and pull them toward the endoscope tip and immobilize the balloon. The sleeve **902** may be a silicone sleeve with a lumen for one or more grasping forceps **900**.

FIGS. **17-21** depict a check valve assembly similar to the check valve **100**. As may be appreciated, the exact dimensions and relationships of the valve and plunger may be altered to achieve optimized alignment, immobilization, and actuation of the check valve. For example, as depicted in the embodiment of FIGS. **17-21**, the spring may have more or fewer coils. The easy grasp loop of the string may be larger or smaller than depicted in FIG. **3**. The string may be formed from multiple strings tied or fastened together. The balloon wall may have a longer stem or a differently contoured outer wall. Other modifications to the balloon, check valve, and catheter are within the scope of this disclosure.

Referring to FIGS. **22-24**, operation of the check valve **100** or check valve of FIGS. **17-21** may be appreciated. For clarity, the catheter **400** is omitted from FIGS. **22-24**, but one may appreciate that the catheter **400** would be aligned with the opening to impact the check valve to effectuate opening of the check valve. In the open position of FIG. **22**, the plunger including plunger head **118**, plunger arm **120**, and plunger tail **122** may be seen extending out of the housing **116**. In this position, the o-ring **128** is not seated in the opening of the housing **116**. The force of the catheter (not depicted) has moved the plunger (including plunger head **118**, plunger arm **120**, and plunger head **122**) toward the interior of the balloon, compressing the spring **124** between the plunger head **118** and the block **130**, and moving the o-ring **128** off of the seat of the opening into housing **116**. Thus, fluid is allowed to flow between the interior of the valve and the interior of the balloon. The fluid may be seen to flow around the string **200**, around the plunger head **118**, plunger arm **120**, and plunger tail **122**, and through the valve opening past the o-ring **128**. It may be appreciated that during inflation, fluid may be introduced into the housing **116** through the endoscope and through catheter (not depicted) to inflate the balloon. During deflation, fluid may be removed in the reverse direction to remove fluid from the interior of the balloon.

With reference to FIG. **23**, the check valve **100** is still in the open position, but is depicted rotated 90 degrees from the position of FIG. **22**. Fluid flow in FIG. **23** flows around the spring **124**, plunger **118**, **120**, **122**, and through the distal end opening of the check valve **100**. Referring now to FIG. **24**, the check valve **100** is depicted in the closed position. As may be appreciated, fluid may be located on the left side of the o-ring **128** (inside the balloon) and on the right side of the o-ring **128** (in the valve housing and in the stomach), however flow between these two areas is not permitted. The spring **124** is shown in the normally extended position, forcing the plunger head **118** toward the right and thus maintaining the o-ring **128** in the closed position.

Referring to FIGS. **25-30**, another valve feature is shown and described. The valve may be similar as the valve previously described. The valve of FIGS. **25-30** may be provided in similar environments such as in a gastrointestinal balloon and/or other remote locations. As previously described, the valve may be in an unstable, remote location (e.g., the gastrointestinal balloon within a patient's stomach). The valve may include a docking station, as previously described, to allow for the immobilization,

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accessing, and opening and closing of the valve. The valve of FIGS. **25-30** may be accessed with a catheter, as previously described. The catheter may be deployed into the docking end or docking station of the valve to allow for access to the interior of the balloon to inflate and/or deflate the balloon. As in previous embodiments, the catheter may track centrally over an attachment (e.g., a string) to allow for the catheter to access the docking end/docking station and subsequently open and/or close the valve.

FIGS. **25-28** show a valve according to another embodiment. The valve **920** may be a valve assembly. The valve **920** and the balloon **924** may have a centerline **26-26**. The valve **920** may include one or more valves **922**. The one or more valves **922** may be arranged in series. That is, a centerline that is coaxial along the axis **26-26** may run through a center of each of the one or more valves **922** such that the one or more valves **922** are coaxial. Although two valves **922a**, **922b** are shown, more or fewer valves may be provided. One valve **922** may be sufficient to allow for the access and docking of the tool within the valve. However, two or more valves **922** may be provided for redundancy, that is, to ensure proper operation of the docking station and back-up sealing between the interior and exterior of a balloon **924**. The valves **922** may be formed as cylindrical discs with an opening therethrough. The opening may be a central opening extending from a top surface to a bottom surface of the valve **922**. The opening may be tapered or conical to facilitate entry of a tool or may be a pinhole opening through the length of the valve that tightly seals around the attachment **928** or tool. The valve **920** may be considered a docking station. That is, the valve **920** may allow for the guiding, alignment, and entering of a tool into a remote area, such as, for example, a gastrointestinal balloon. As discussed previously, the remote location may be mobile, for example, the floating gastrointestinal balloon within the stomach of a patient. The mobility of the remote location may make the valve **920** difficult or impossible to access. Such mobility may require, in conventional valves, the removal of the valve from the remote location to provide a stable environment for accessing the valve. The valve **920** may provide a docking station that allows for the immobilization and subsequent inflation/deflation of the balloon while the balloon remains in the remote, mobile or floating location. That is, the valve **920** allows for the docking, immobilization, and opening and/or closing while the gastrointestinal balloon remains in the patient's stomach.

The valve **920** may be coupled to the balloon **924**, such as previously described. The balloon **924** may have an opening **926** for receiving the valve **920**. As shown and described with respect to previous embodiments, the opening **926** may be located within the balloon **924**. That is, the opening **926** may not extend past an exterior surface **924a** of the balloon **924**. The opening **926** may have an end **926a** that extends downward from the exterior surface **924a** and into the interior of the balloon. The balloon **924** is shown truncated for clarity in FIGS. **25** and **26**. However, the balloon **924** may extend in a manner that creates a cavity that is accessible through the valve **920**. The opening **926** may be substantially cylindrical such that the opening **926** is a cylindrical bore surrounded by a wall of the balloon **924**. The opening **926** may house the one or more valves **922** and the one or more spacers.

The valve **920** may include an attachment **928**, such as an elongated member, such as a string that may have a connector **930**. The connector **930** may be an "easy grasp" loop that exits into the exterior space (e.g., the space surrounding the balloon) of the balloon **924**. The attachment

928 with easy grasp loop or connector 930 may be formed of nylon. The attachment 928 may be an elongated member, a bungee, a stretchy member, a string, a cable, a cord, other elongated device, or anything in accordance with the principles of invention capable of coupling to the valve 920 and being grasped by a tool, such as described with respect to previous embodiments. For ease of description, the attachment 928 is described as a string and the connector 930 is described as a loop, however other modes are contemplated. The string or attachment 928 may be a single string having opposing distal ends 928a, 928b (FIG. 27), each coupled to a block 932. The block 932 may be a beam or bar to which the distal ends 928a, 928b are fixed or coupled. Alternatively, the block 932 may be the distal ends 928a, 928b fixed together such that they may not pass through an opening in the spacer 934. The block 932 may form a T-shape (for example, see FIG. 27) such that the string or attachment 928 may be prohibited or preventing from detaching or uncoupling from the valve 920. Although the block 932 is shown in a left to right configuration in FIG. 25, the block 932 may be orientated along any axis of the valve 920 and/or balloon 924 so long as the block 932 prevents removal of the attachment 928 from the valve 920. The loop or connector 930 is shown in an enlarged circular shape in FIGS. 25-28, however the loop or connector 930 may take a smaller shape, such as described with respect to FIG. 3.

As mentioned, the valve 920 may include one or more valves 922. The valve 920 may include one or more spacers 934, 936, 938. A first spacer 934 may be located at a first end of the valve 920. A second spacer 936 may be located on the first spacer 934, as shown in FIG. 26. A first valve 922a may be located on the first spacer 934. A third spacer 938 may be located on the first valve 922a. A second valve 922b may be located on the third spacer 938. Although shown as separate components one or more of the one or more valves 922 and/or the spacers 934, 936, 938 may be formed integrally and/or coupled together. That is, for example, the first spacer 934 and the second spacer 936 may be formed as a single, integral component.

Each of the one or more valves 922 and the one or more spacers 934, 936, 938 may have an opening extending therethrough. For example, the first valve 922a and the second valve 922 may have openings 940a, 940b, respectively, extending through the valves. The one or more spacers 934, 936, 938 may have an opening 934a, 936a, 938a, respectively, extending through the spacers. The openings 940a, 940b, 934a, 936a, 938a may be coaxial and/or aligned. The attachment 928 may extend through the openings 940a, 940b, 934a, 936a, 938a and be coupled to or integral with the block 932. That is, the first distal end 928a and the second 928b may be coupled to, integral with, or otherwise attached to the block 932. The attachment 928 may be extended through the openings of the one or more valves and the one or more spacers such that the loop or connector 930 extends out of the opening 926 and into the exterior space around the balloon 924.

The opening 934a may be sized such that the attachment 928 may extend therethrough and also such that the block 932 is prohibited from extending through the opening 934a. In this manner, the block 932 may anchor or hold the attachment 928 to the valve 920. The openings 936a and/or 938a may be formed of any size that is sufficient to allow passage of the attachment 928 and the tool, to be described later.

The openings 940a, 940b of the one or more valves 922 may be formed of a size just large enough to accommodate the attachment 928. That is, the attachment 928 may be

bounded on all sides by walls of the valve 922 such that no air or other fluid may flow through the one or more valves 922. The one or more valves 922 may be formed of a flexible material such that, upon insertion of a tool, the openings 940a, 940b may be enlarged to allow passage of the tool. The one or more valves 922 may be formed of a solid silicone. The one or more valves 922 may be a self-sealing valve. The self-sealing valve may be formed of a self-sealing material, such as, for example, silicone. The one or more valves 922 may be formed of a flexible material, such as solid silicone, with a small hole or opening at or around the center of the valve. That is, the one or more valves 922 may be formed of a material that is allowed to be opened or expanded from a first state (closed) to a second state (open) and back to the first state (closed). The one or more valves 922 may be formed of a material that allows for repetitive opening and closing. The openings 940a, 940b may be in the normally closed position, the position where the attachment 928 extends therethrough but no fluid is permitted to pass through. The openings 940a, 940b may be transitioned from the normally closed position to the open position by advancing a tool through the opening 940a, 940b to expand, distort or otherwise flex the openings 940a, 940b to accommodate the tool.

As mentioned, the valves 922 may be formed of a flexible material with a tiny, pinhole sized hole extending there-through. The pinhole may be the opening 940 and may be sized to accommodate the attachment 928 in the closed position but also prevent flow therethrough. In the open position, the pinhole may be able to accommodate both the attachment 928 and the tool. The opening 940 may tightly surround the attachment 928 in the closed position and may tightly surround the tool in the open position. The opening may be a pinhole opening that runs the length of the flexible (e.g., silicone) valve 922. The valve 922 may be a loose duck-valve or duck-valve like valve.

Thus, in a first state, the openings 940a, 940b may have a first size sufficient to allow passage of the attachment 928 but not sufficient to allow passage of fluids and/or passage of a tool with a diameter larger than the attachment. In a second state, the openings 940a, 940b may have a second size sufficient to allow passage of the attachment 928 and sufficient to allow passage of the tool with a diameter larger than the attachment. In the second state, a fluid may be passed through tool and thus through the one or more valves 922. The fluid may exit through ports 942 in the first spacer 934 and allowed to flow into the interior of the balloon 924. Although 8 ports 942 are shown in FIG. 28, more or fewer may be provided. As many ports 942 as possible may be provided in the spacer 934 such that the flow through the spacer 934 from the catheter may be as great as possible without reducing the integrity of the valve 920 and/or without allowing the attachment 928 to disengage from the valve 920. Additionally, the ports 942 may take alternative shapes or sizes. For example, two c-shaped ports may be provided or a single, annular or donut shape port may be provided.

In an embodiment where a single valve 922 is present, the spacer 938 and/or the spacer 936 may be omitted. In an embodiment where three valves 922 are present, an additional spacer may be located between the second valve 922b and a third valve (not depicted) about the second valve 922b. In the case where three or more valves 922 are present, the opening 926 and the associated walls of the balloon 924 may be extended further into the interior of the balloon 924 such

that the entirety of the valve **920**, including all valves **922** and all spacers, are located below the surface **924a** of the balloon **924**.

In an embodiment, the valve **920** may include a second block **946**, as shown in FIGS. **29** and **30**. The second block **946** may assist in preventing removal of the attachment **928** from the valve **920**. The second block **946** may be substantially spherical. Alternatively, the second block **946** may be T-shaped like the block **932**. The block **932** and the second block **946** may hold the attachment **928** in the opening **934a** and prevent removal therefrom. The second block **946** may prevent the attachment **928** from being pushed into the valve **920** and/or the interior of the balloon **924**. The valve **920** and balloon **924** may have a centerline **29-29**. The assembly may be aligned along the axis **29-29** similar to the assembly of FIGS. **25-28** being aligned on the axis **26-26**.

The valve **920** of FIGS. **25-28** and/or FIGS. **29-30** may be overmolded such that the valve **920** takes the appearance or may be considered a single, integral part. The valve **920** may then be inserted into the balloon **924** as a single component. The second valve **922b** may be redundant of the first valve **922a** and may allow for confirmation and/or validation of closing and opening of the valve.

The valves may be separated by one or more spacers. One or more of the spacers may be optional. One or more spacers may be omitted. That is, the valves may be separated by one or more spacers, may not be separated by one or more spacers, or combinations thereof. The valves may be next to each other, together in sequence, next or near in line or sequence, adjacent, or contiguous. The valves may abut one another. That is, the bottom surface of one valve may touch or rest on the top surface of a subsequent valve. Although described with two spacers, more or fewer may be provided. Although described with two valves, more or fewer may be provided.

The valve **920** may operate in a manner similar to the method described with respect to FIG. **9**. During operation, an endoscope may be extended into a patient's stomach. A catheter **400** (FIG. **9**) with a grasping tool **500** may be inserted therethrough. The grasping tool **500**, such as grasping forceps used in endoscopic procedures may be inserted into the lumen **404** of the catheter **400** through the homeostatic valve **408** (FIG. **8**). During the procedure, the balloon, valve, catheter, and grasping tool may all be located within the patient's stomach. The forceps or similar grasping tool **500** may grasp the string or attachment **928** by the easy grasp loop or connector **930** and pull the valve **920** toward the catheter **400**. At the same time, the catheter **400** may be advanced toward the valve **920**. This may easily guide the grasping tool **500** to the target or docking end, the opening **940b** of the second valve **922b**. The catheter **400** then enters the opening **940b**, passes through the openings **938a**, **940a**, and **936a**. The tip **402** (FIG. **9**) of the catheter **400** may stop or be located in a space **944** of the spacer **934**. The tip **402** may be offset from the inner surface of the spacer **934** such that fluid may easily flow from the lumen **404** of the catheter out of the ports **942** in the spacer **934**. Such alignment of the catheter **400** with the check valve **100** allows for both immobilization of the valve **920** and thus the balloon **924** and entry into the valve **920** and balloon **924**.

While fixating the catheter **400** and gastroscope with one hand, the grasping tool **500** (now grasping string or attachment **928**) within the catheter **400** is pulled back with the other hand. This motion causes the tip **402** to extend through the opening **940b**. Continued pulling of the string or attachment **928** pushes the catheter tip **402** into the remaining openings and finally into the space **944**. Alternatively, once

the catheter **400** is aligned with the docking end, the catheter **400** and thus the catheter tip **402** may be pushed into the openings **940b** and then continue to be pushed through opening **940a** and into the space **944**. The opening **940b** and/or the opening **940a** may be tapered to assist and ease guiding of the catheter **400** into the valve **920**. So long as the catheter **400** extends through the valve **920**, fluid may be allowed to flow into and/or out of (e.g. bi-directional flow) the balloon **924**. To close the valve **920**, the catheter **400** may be removed, thus returning the openings **940a**, **940b** back to the first state.

After inflation or deflation is completed, the catheter **400** may be removed from the docking station or entry to the valve **920**. The tool **500**, catheter **400**, and endoscope may be removed from the patient's stomach and from the patient.

During the passage of the catheter **400** through the openings of the valves **922** and spacers, the catheter **400** may track over the attachment **928** which also extends through the openings of the valves **922** and spacers. That is, in the position where the valve is open and the catheter is in place for inflation and/or deflation of the balloon **924**, both the catheter **400** and the attachment **928** may extend through the openings **904a**, **940b**, **938a**, **936a**, and **944**. The openings **938a**, **936a**, and **944** may be sized to accommodate the catheter **400**. The openings **940a**, **940b** may be formed in a self-sealing material such that the openings are permitted to expand to a size accommodating the catheter **400** and then seal or contract to a position prohibiting entry of the catheter **400**. In the position where the valve is closed and the catheter is removed, the attachment **928** may extend through the openings **904a**, **940b**, **938a**, **936a**, and **944**, but fluid flow through the valve **920** and specifically valves **922** may be prohibited.

Although the catheter **400** is described as impacting the valve **920**, the catheter may be forced into the valve in other manners. For example, the catheter **400** may be secured to the opening **940a** with a turning motion. That is, the catheter **400** may have a tip **402** which may be screwed or threaded into the valve opening **940a**. One may appreciate that the inner surface of opening **940a** and the outer surface of the catheter tip **402** may have complementary threads or other surfaces to allow the rotational and longitudinal movement.

Accordingly, any of the aforementioned valves and/or assemblies may allow for tracking of a tool, such as a catheter, over the attachment to provide guidance, alignment, and entry of the tool into the valve. That is, the attachment may be grasped by a first tool, and a second tool, such as a catheter, may be guided or tracked over the attachment to allow entry of the second tool into the opening of the balloon, thus actuating or opening the valve located in the opening of the balloon. In this manner, the attachment may operate as a guide or tracking member to allow for access and opening of the valve. The attachment may extend through the center of the valve or valves. The catheter may guide centrally over the attachment such that the catheter enters centrally into the opening of the balloon and/or the valve.

The valve of the present disclosure may be a valve having an attachment (e.g., a string) extending centrally there-through. That is, the valve may have the attachment running or extending through the venter of the valve. The valve may be two contiguous valves with an attachment or string that runs or extends through their center and exits the balloon. The attachment or string may enable a user to grasp the attachment or string and use the attachment or string as a tract to guide the catheter or other tool directly into the

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valve. The central string may facilitate entering the valve and changing (e.g., inflating or deflating) the balloon volume.

As may be appreciated from the foregoing disclosure, the check valve of the disclosure may be used in a gastrointestinal balloon. The disclosure further includes a device, system, and method to immobilize the balloon within the body while at the same time providing access to the valve in such a way that the volume can be adjusted. The system can include the balloon and check valve, the catheter, and tool that is configured for grasping onto part of the valve to hold it in place. The disclosure may also include the method of performing this maneuver.

Additionally, the valve may be a check valve, although other valve types (e.g. ball valves, sleeve valves, etc.) may be provided. The check valve may include a spring inside that maintains the valve in a closed position and the spring is compressed to open the valve.

The check valve and balloon may be suitably immobilized while a force is applied to compress the spring. The balloon may be used in the stomach, but may be used for any inflatable, implanted device. The check valve may include a portion that may be grasped by a tool inserted to inflate/deflate the balloon. In some embodiments, this is a loop that extends out from the valve and is grasped by a grasping tool. The loop may be pulled (away from the balloon) by the catheter/device to immobilize the balloon. Grasping forceps, inserted through the catheter, may be used to pull the loop. After the catheter engages with the valve, the loop can optionally be released with the catheter remains engaged to inflate/deflate the balloon.

Additionally, as described, the check valve may be used in systems, methods, and environments other than a gastrointestinal balloon. The check valve in may have a string, or other extension, extending from the check valve. The string/extension may be grabbed by a tool inserted into the flow path to access the check valve.

Although the foregoing description is directed to the preferred embodiments of the invention, it is noted that other variations and modifications will be apparent to those skilled in the art, and may be made without departing from the spirit or scope of the invention. Moreover, features described in connection with one embodiment of the invention may be used in conjunction with other embodiments, even if not explicitly stated above.

The invention claimed is:

1. A valve assembly for a gastrointestinal balloon, the valve assembly comprising:

a valve formed of a self-sealing material, the valve having a first end, and a second end, and a valve opening extending from the first end to the second end;

a spacer located at the first end of the valve, the spacer having a first spacer opening and at least one second spacer opening; and

an attachment configured to extend through the first spacer opening and the valve opening, the attachment having an anchor configured to be located within the gastrointestinal balloon and configured to prevent removal of the attachment from the first spacer opening and the valve opening,

wherein, when in an open condition, the valve is configured to allow fluid flow through the at least one second spacer opening and, in a closed condition, the valve is configured to prevent fluid flow through the at least one second spacer opening.

2. The valve assembly of claim 1, wherein the valve is a first valve, the valve assembly further comprising a second

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valve located in series with the first valve, the second valve formed of a self-sealing material, wherein the attachment extends through the second valve.

3. The valve assembly of claim 2, wherein the spacer is a first spacer, the valve assembly further comprising a second spacer located in series and between the first valve and the second valve.

4. The valve assembly of claim 2, wherein the valve opening through the first valve is a first opening through the first valve and further comprising a second opening through the second valve, further wherein the first opening through the first valve and the second opening through the second valve are aligned.

5. The valve assembly of claim 4, wherein the first opening through the first valve and the second opening through the second valve each have a first state preventing fluid flow therethrough and a second state allowing passage of a tool.

6. The valve assembly of claim 5, wherein, in the first state, the first opening through the first valve and the second opening through the second valve are closed around the attachment.

7. The valve assembly of claim 1, wherein the spacer is configured to separate the valve from an interior of the gastrointestinal balloon.

8. The valve assembly of claim 1, wherein the self-sealing material is silicone.

9. The valve assembly of claim 1, wherein the valve is formed as a cylindrical disc with the valve opening therethrough, the valve opening configured to receive the attachment.

10. The valve assembly of claim 1, further comprising an open position and a closed position, wherein the valve is configured to move to the open position in a stomach.

11. The valve assembly of claim 10, wherein the open position permits passage of a tool and the closed position prohibits flow of fluid.

12. The valve assembly of claim 1, wherein the anchor is a block configured to prevent removal of the attachment from the valve and the spacer.

13. The valve assembly of claim 1, wherein the attachment extends through a center of the valve.

14. A method of accessing a gastrointestinal balloon, the method comprising:

grasping an attachment extending through a self-sealing valve;

tracking a tool over the attachment and toward the self-sealing valve;

pulling the self-sealing valve toward the tool simultaneously with the tracking;

pushing the tool through an opening in the self-sealing valve, still tracking over the attachment, thus opening the self-sealing valve; and

inflating or deflating the gastrointestinal balloon through the tool.

15. The method of claim 14, wherein tracking the tool over the attachment comprises passing the attachment through a lumen of the tool.

16. The method of claim 14, wherein pushing the tool through an opening in the self-sealing valve comprises expanding the opening by entering a distal end of the tool into the opening.

17. The method of claim 14, wherein the self-sealing valve is a series of two self-sealing valves formed of silicone.

18. The method of claim 14, further comprising removing the tool from the self-sealing valve thus closing the self-sealing valve.

19. A system for accessing an interior of a gastrointestinal balloon, the system comprising:

a gastrointestinal balloon;

two valves having coaxially aligned, tapered openings;

and

a string running through a center of each of the two valves and exiting the gastrointestinal balloon; and

an anchor configured to permanently hold the string in the two valves, the anchor located within the gastrointestinal balloon,

wherein the string is configured to be grasped by a user to track a tool over the string and directly through the two valves and into the interior of the gastrointestinal balloon and wherein the anchor is fixed with respect to the two valves.

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